

Teaching Dossier

Arman Jahangiri

Email: arman77@uw.edu

Website: armanjg.github.io

LinkedIn: linkedin.com/in/armanjg

This dossier presents an evidence-based account of my teaching in mathematical and statistical sciences. It is supported by course materials, student evaluations, supervisory observations, and selected instructional artifacts.

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Summary of Teaching Profile

I teach mathematics and statistics with the goal of helping students become confident, independent, and conceptually grounded learners. My teaching focuses on developing students' ability to communicate mathematical ideas clearly and connect to the underlying structures that justify them. Across university teaching, tutoring, and enrichment programs, I have emphasized active engagement, transparent expectations, and structured opportunities for students to explain, test, and refine their thinking.

At the University of Washington, I have served as an instructor of record and a teaching assistant. In Summer 2026, I taught MATH/STAT 394: Probability I as the instructor of record. I was responsible for the design and delivery of the course, including lectures, homework, examinations, grading policies, student communication, accessibility arrangements, review materials, and the [public course website](#) that I hosted on my GitHub for a more organized and self-contained delivery, in addition to using the Canvas platform.

I have taught sections in various topics such as algebra, precalculus, the full calculus sequence, and upper-division mathematics, including Linear Optimization at the University of Washington, and graduate and undergraduate courses in statistics and data science, such as Theoretical Statistics and Data Analysis at the University of Calgary.

Beyond formal university instruction, I have worked extensively as a private mathematics and statistics tutor and as an instructor in mathematics enrichment programs.

A central feature of my teaching is the deliberate integration of conceptual understanding with structured practice. I aim to design learning experiences that help students interpret ideas meaningfully, recognize assumptions, justify methods, and adapt familiar tools to new settings. I also prioritize creating classroom environments in which students feel comfortable asking questions, revising their thinking, and participating actively in the learning process.

This dossier documents my teaching philosophy, instructional practices, independent course design, student support, evidence of teaching effectiveness, and continuing professional development as an educator.

Teaching Philosophy

Teaching across educational frameworks

My approach to teaching has been shaped by experiences across multiple countries, languages, and educational systems. These experiences have convinced me that the quality of learning increases substantially when students move beyond passive observation. They develop stronger understanding when they articulate their reasoning, receive timely feedback, and have structured opportunities to refine their thinking.

As a first-generation college graduate, I am aware that success in mathematics depends on both mathematical ability and familiarity with academic expectations that are sometimes left implicit. Students may be uncertain about how to ask productive questions, approach an unfamiliar problem,

interpret feedback, or evaluate their own understanding. I therefore try to make expectations visible and to create learning environments that support intellectual risk-taking and gradual development of independence.

My goal as an educator is to cultivate students' ability to think with clarity, rigor, and independence. I support students to learn how to communicate complex ideas precisely, construct and evaluate arguments, identify assumptions, and connect individual mathematical techniques to broader theoretical and applied questions.

Externalizing reasoning as a foundation of learning

I believe students learn mathematics most effectively when they externalize. A correct answer is useful evidence of learning, but the reasoning that produced it is usually more informative. Students deepen their understanding when they explain why a method applies, compare alternative approaches, identify assumptions, and analyze mistakes.

In my classroom, I therefore encourage students to make their thinking visible through discussion, short written explanations, polls, guided problem solving, and questions during proofs and examples. I emphasize mathematical communication and interpretation rather than presenting mathematics as a sequence of formulas to reproduce.

My teaching experiences have repeatedly shown me that students can become proficient at executing procedures while remaining uncertain about why those procedures work, when they apply, or how they should change when the structure of a problem changes. For that reason, I view procedural fluency and conceptual understanding as complementary rather than competing goals.

Structured active learning

I incorporate active-learning strategies that require students to make mathematical decisions during class. The goal is not activity for its own sake, but opportunities for students to retrieve knowledge, test understanding, and reveal misconceptions while learning is still taking place.

Depending on the course and topic, I ask students to:

- think about the form or sketch of an answer before carrying out a calculation - an outlook to the full solution before writing everything down;
- identify which technique is appropriate and explain their decision;
- compare solution strategies with classmates;
- diagnose errors in proposed arguments or calculations;
- interpret a mathematical result verbally or graphically;
- apply a recently introduced idea independently before seeing a complete solution.

These activities help me identify misconceptions early and give students practice making the kinds of decisions that independent problem solving requires.

I also try to normalize uncertainty and revision. Students should be able to propose an idea, discover

why, if it does not work, and use that failure productively. Mistakes can provide unusually clear evidence about the structure of a concept when they are analyzed rather than simply corrected.

Conceptual understanding through multiple representations

Students can easily experience mathematics as a collection of isolated formulas and procedures, particularly in introductory courses. I therefore use multiple representations whenever they clarify the underlying structure of an idea: symbolic, graphical, numerical, verbal, or computational.

In probability, for example, a distribution can be approached through its formula, its graph, its probabilistic interpretation, its moments, its relationship to other distributions, and simulations illustrating its behavior. Moving among these representations helps students understand what mathematical objects mean rather than only how to manipulate them.

My international teaching experience has reinforced the importance of this flexibility. Students bring different educational backgrounds, intuitions, and problem-solving habits into the classroom. Providing more than one entry point into an idea broadens access without reducing mathematical rigor.

Inclusive teaching and transparent expectations

I try to ensure that success in my courses does not depend on students already knowing the hidden conventions of university mathematics. I therefore make expectations explicit through learning goals, review materials, sample problems, clearly stated policies, and transparent assessment structures.

This approach is especially important in courses containing students with different levels of preparation. Transparency does not mean making a course less challenging. Rather, it means allowing students to direct their effort toward learning the intended material instead of guessing what the instructor expects.

I also provide multiple ways for students to access course materials and seek help. For example, by providing course materials through a public website in both PDF and accessible HTML formats, when the official platforms make accessibility hard, while private components of the course such as grades and course administration remained accessible through Canvas.

Teaching Experience

Institution	Role	Course	Term	Scale
University of Washington	Predoctoral Instructor Instructor of Record	Probability I <i>MATH/STAT 394</i>	Summer 2026	40
	Predoctoral Teaching Associate	Algebra with Applications <i>MATH 111</i>	Spring 2026	40 40
	Predoctoral Teaching Associate	Calculus with Analytic Geometry II <i>MATH 125</i>	Winter 2026	30; 30
	Predoctoral Teaching Associate	Precalculus <i>MATH 120</i>	Autumn 2025	30; 30
	Predoctoral Teaching Associate	Linear Optimization <i>MATH 407</i>	Summer 2025	40
	Predoctoral Teaching Associate	Calculus with Analytic Geometry III <i>MATH 126</i>	Spring 2025	40 40
	Predoctoral Teaching Associate	Calculus with Analytic Geometry II <i>MATH 125</i>	Winter 2025	30 30
	Predoctoral Teaching Associate	Calculus with Analytic Geometry I <i>MATH 124</i>	Autumn 2024	30 30
University of Calgary	Graduate Teaching Assistant	Statistical Methods in Data Science <i>DATA 606</i>	Winter 2024	—
	Graduate Teaching Assistant	Introduction to Statistical Inquiry <i>STAT 213</i>	Winter 2024	—
	Graduate Teaching Assistant	Introduction to Statistics II <i>STAT 217</i>	Winter 2024	—
	Graduate Teaching Assistant	Statistical Data Analysis <i>DATA 602</i>	Autumn 2023	—
	Graduate Teaching Assistant	Introduction to Statistics I <i>STAT 205</i>	Autumn 2023	—
	Graduate Teaching Assistant	Introduction to Theoretical Statistics <i>STAT 323</i>	Summer 2023	—
	Graduate Teaching Assistant	Linear Methods I <i>MATH 211</i>	Winter 2023	—
	Graduate Teaching Assistant	University Calculus I <i>MATH 249/265</i>	Winter 2023	—
	Graduate Teaching Assistant	Mathematics / Statistics	2022–2024	—
MATH Pro (Canada)	Tutor	Mathematics, Probability, and Statistics	2023–2024	50+

Institution	Role	Course	Term	Scale
Private / Online	Private Tutor	Mathematics and Statistics	2022–2024	50+

Course Descriptions

The following summaries provide context for selected courses in which I have served as instructor of record, Predoctoral Teaching Associate, or Graduate Teaching Assistant.

University of Washington

MATH/STAT 394: Probability I An introduction to probability covering foundational probability rules, conditional probability, independence, random variables, expectation and variance, discrete and continuous distributions, transformations, joint distributions, probabilistic limit ideas, and modes of convergence. The course emphasizes both mathematical reasoning and the interpretation of probabilistic models.

MATH 111: Algebra with Applications An introductory quantitative-reasoning course emphasizing algebraic modeling, graphs, exponential and logarithmic functions, and applications arising in business and economics.

MATH 120: Precalculus A preparatory course for the calculus sequence covering functions, trigonometry, exponential and logarithmic models, polynomial and rational functions, and foundational analytical skills.

MATH 124: Calculus with Analytic Geometry I The first course in the standard calculus sequence, focusing on limits, continuity, differentiation, optimization, and applications of derivatives.

MATH 125: Calculus with Analytic Geometry II The second course in the calculus sequence, emphasizing integration, the Fundamental Theorem of Calculus, applications of integration, and sequences and series.

MATH 126: Calculus with Analytic Geometry III The third course in the calculus sequence, including Taylor expansions, vectors and geometry in three dimensions, partial derivatives, multiple integration, and multivariable applications.

MATH 407: Linear Optimization An upper-division course in linear optimization, including linear programming, convexity, duality, simplex-type methods, and mathematical modeling.

Independent Course Design:

Course context and instructional responsibility

In Summer 2026, I served as instructor of record for **MATH/STAT 394: Probability I** at the University of Washington. I was responsible for making course-level decisions about content creation and organization, assessment, student support, communication, accessibility, and the relationship among lectures, homework, review materials, and examinations. The course was taught in a hybrid format. I developed and maintained a public course website containing the syllabus, homework assignments and solutions, lecture slides, annotated lecture materials, supplementary notes, exam materials, interactive probability resources, and an optional project.

Public and accessible course infrastructure

A major goal in designing the course website was to reduce unnecessary organizational barriers. Students should be able to locate materials easily and understand how different resources fit together.

I therefore maintained a dedicated public course website, covering all the required material students need:

armanjg.github.io/MATH-394-Probability-1-v2/

The site included homework assignments, their solutions, midterm and final examination materials and their solutions, practice and review resources, question banks, practice exams, lecture slides, annotated lecture slides, recorded lecture videos, supplementary probability notes, interactive visualizations, and the project guide.

Curriculum organization

I designed the course so that new probability concepts were connected explicitly to earlier ideas. For example, when introducing the exponential distribution, I connected it to the previously studied geometric distribution. The comparison provided an opportunity to distinguish continuous and discrete waiting-time models while reinforcing the broader conceptual role of memorylessness. Similarly, new continuous distributions were used as opportunities to revisit probability density functions, normalization, expectation, and variance. Later topics therefore functioned not only as new material but also as structured retrieval of earlier concepts. The course moved among mathematical derivation, probabilistic interpretation, visualization, and application. My aim was for students to understand both how probability calculations are performed and what the resulting quantities mean; see the [Course Syllabus](#) for a detailed topics' schedule.

Homework and structured practice

The weekly homework sequence served several purposes: regular practice, development of mathematical reasoning, preparation for examinations, and exposure to problems requiring students to combine ideas rather than reproduce a single classroom example. Detailed solutions were released after the deadlines to allow students to revisit their reasoning. These solutions were written as instructional

documents rather than answer keys, with mathematical steps and explanations presented in a form students could use for later review.

The [first homework set](#) and [final homework set](#) are included in the appendix as examples of the development of the course from foundational probability ideas to later topics.

Assessment design

Assessment was organized around [homework](#), a [midterm examination](#), a cumulative [final examination](#), an optional [mini-project](#), and active classroom participation through [Ed Discussion](#) questions. Detailed [midterm](#) and [final solutions](#) were provided to clarify grading expectations and enhance students' understanding of the material.

The midterm and final were designed to evaluate more than formula recall. Questions required students to identify appropriate probabilistic structures, carry out mathematical calculations, and interpret results.

Substantial assessment preparation was provided including [midterm practice](#) that simulated the midterm, along with [solutions](#), a [midterm question bank](#), and a [final question bank](#). The goal was to make the learning targets transparent without making the assessments mechanical.

The purpose of the project was to create an additional opportunity for students to engage with probability beyond the standard textbook structure. It provided a different mode of mathematical engagement and allowed students to conduct research in and communicate probability.

Technology and interactive resources

Technology was used when it served a clear pedagogical purpose.

During lectures, I used Poll Everywhere for low-stakes checks of understanding. For example, after learning a new topic, students could attempt a related question independently before we discussed the answer, and check their correctness unanimously. This provided immediate information about student understanding. When responses were divided between correct and incorrect answers, I could ask students to diagnose whether the difficulty arose from algebraic manipulation or from the probability concept itself.

I also developed interactive resources illustrating probabilistic ideas, including [properties of cumulative distribution functions](#), [normal approximations](#), [bivariate normal distributions](#), and [convergence of sample means for several distributions](#).

These visual and computational representations complemented the symbolic development presented in lecture.

Hybrid instruction

Teaching a hybrid course required attention to students participating through different modalities. I maintained digital access to lecture materials so that students could revisit course content outside the classroom. Students attending remotely could also participate through Zoom. The experience

reinforced the importance of designing activities that do not unintentionally privilege only the students physically closest to the instructor.

Instructional Practices and Pedagogical Strategies

Connecting new ideas to prior knowledge

I regularly introduce new concepts by asking students to retrieve something they already know. This practice has two purposes. First, it makes prior knowledge available when students need it. Second, it helps students organize mathematics as a connected system rather than a sequence of isolated topics.

Checks for understanding

I use questions, polls, short independent attempts, and student explanations to obtain evidence about understanding during instruction.

A particularly useful pattern is:

1. introduce or derive an idea;
2. work through an example;
3. ask students to apply a closely related idea independently;
4. inspect the distribution of responses;
5. discuss the reasoning behind both correct and incorrect approaches.

The value of the final step is especially important. A wrong answer can arise from a conceptual misunderstanding, an algebraic error, a notation problem, or a misreading of the question. Distinguishing among these possibilities produces more useful feedback than simply displaying the correct answer.

Building supportive learning environments

Mathematics courses can produce substantial anxiety and self-doubt. I therefore try to create a classroom culture in which asking questions, revising one's thinking, and learning through mistakes are treated as normal parts of mathematical work. I also recognize that students learn at different paces and benefit from encountering material through multiple modalities. Students may differ in how they process auditory, visual, verbal, and experiential information, as well as in their prior knowledge and cognitive strategies. Rather than relying on a single mode of instruction, my courses use a multimodal and active-learning approach that combines guided examples, conceptual discussion, written explanation, visualization, individual practice, and anonymous polling. This variety helps reduce cognitive barriers and gives students multiple opportunities to encode, process, and retrieve important concepts.

Assessment, Feedback, and Academic Integrity

Assessment philosophy

Assessment should reward the forms of thinking that a course claims to value.

In my assessment design and grading, I emphasize:

- **Correctness:** reasoning and conclusions should be accurate;
- **Reasoning:** students should show why a method or result is justified;
- **Method selection:** students should recognize which tool is appropriate;
- **Communication:** notation and logical structure should make reasoning understandable;
- **Interpretation:** students should understand what an answer means in context.

Feedback that students can use

Feedback is most useful when students can translate it into a specific future action.

When possible, my feedback therefore identifies:

- something in the student's reasoning that should be retained;
- the highest-leverage correction or improvement;
- a specific next step for practice or review.

I also use class-wide feedback when several students encounter the same difficulty. In such cases, an error pattern becomes evidence about instruction, not merely about individual performance.

Academic integrity and productive collaboration

I distinguish explicitly between productive collaboration and outsourcing mathematical thinking.

Students benefit from discussing concepts, comparing strategies, and asking questions. At the same time, submitted work should represent the student's own understanding and should comply with the specific collaboration rules of the assignment.

My broader goal is to frame academic integrity as part of learning rather than only as a disciplinary rule: students should use outside help in ways that increase, rather than replace, their ability to reason independently.

Teaching Effectiveness

Student evaluations across courses

Student feedback across my teaching appointments has repeatedly highlighted conceptual explanation, accessibility, responsiveness to questions, and the use of structured problem solving.

Representative comments from earlier University of Washington courses include:

“Arman’s advanced understanding of mathematics gave a broader point of view towards the material comparatively to the curriculum itself.”

“The content actually made more sense when he would break it down into different steps to further expand your thinking.”

“Arman was really helpful in answering my questions. He was able to explain complex problems really well. What I learned during the quiz section helped me perform better on the exams.”

“Arman was great at explaining the math material and was always willing to go beyond simply teaching what he has to teach and allowing us to understand the material better by exploring the concepts clearer.”

Students have also emphasized the value of support outside scheduled class time:

“Definitely the extra review material that he provided along with the Zoom meetings and extra review time for exams.”

“I think the time given to ask questions about the homework contributed the most to my learning because that was what gave me the most clear understanding on what I was being asked to do.”

Complete evaluation reports are included in the appendix [[Teaching Evaluation Reports](#)].

MATH/STAT 394 Probability I: Summer 2026 student feedback

The course enrolled 21 students, of whom 14 completed the evaluation, for a response rate of approximately 67%. The combined overall median was 4.1 out of 5, while the median rating for instructor contribution was 4.5 out of 5. Teaching effectiveness received a median of 4.0 out of 5. Students also rated the use of examples highly, with a median of 4.7 out of 5, and assessment and grading received a median of 4.6 out of 5. The Challenge and Engagement Index was 5.0 out of 7 [[Official Student Evaluation](#)].

Written feedback reinforced several themes that were also visible in the numerical evaluation. Students particularly valued the organization and availability of course materials, including the website, slides, annotated slides, recordings, visualizations, homework solutions, and exam-preparation resources.

At the same time, student feedback identified an important area for improvement. Some students perceived a gap between the difficulty of examples developed in lecture and the most challenging

homework problems. I view this as useful evidence for future course design. A revision of the course should preserve challenging homework while providing a more explicit bridge between foundational examples and problems requiring several ideas to be combined. Possible revisions include intermediate “bridge” problems, shorter in-class challenge problems, and more explicit discussion of how familiar techniques are assembled in unfamiliar settings.

Supervisory and mentor observations

My teaching has been observed at multiple stages of my work. They identify several recurring strengths in my teaching: clear mathematical communication, careful preparation, organization, a calm and approachable classroom presence, and deliberate attention to student understanding.

In an early observation of my calculus teaching, Clare Minnerath emphasized both clarity and classroom presence. They noted that my board work was organized, my voice was clear, and that I responded constructively to student questions and incomplete answers.

Her overall assessment was that “Arman appeared confident and experienced at the board.”

A later observation by Dr. Andrew Loveless similarly emphasized approachability, organization, and student engagement. He noted that I circulated throughout the room, spoke individually with students, and created an environment in which students were actively discussing mathematics with one another. He described the atmosphere as calm and low-pressure and observed that my explanations at the board were logically organized and easy to follow. As he wrote, “Students were engaged and working diligently,” and concluded that “You are building a solid classroom presence.”

The observation of my Probability I course identified many of the same qualities in the context of a course for which I had full instructional responsibility. The report described the lecture as “effective and carefully prepared” and emphasized the strength of the course infrastructure, including the organized course website, review materials, posted solutions, and lecture notes. It also noted that my explanations were “calm, well paced, and well matched to the course level.” Particular attention was given to the way I connected new material to previous ideas, for example by relating the exponential distribution to the geometric distribution and by revisiting the definition of a probability density function while introducing new content.

The report also highlighted my use of formative assessment. A Poll Everywhere activity asked students to apply a recently introduced idea independently, after which I used the distribution of responses to distinguish conceptual misunderstandings from algebraic errors. The observer described this as a particularly effective instructional choice because it made student thinking visible and created an opportunity to respond directly to the source of students’ difficulties. The learning environment was characterized as “calm, supportive, and professionally organized,” with students asking thoughtful questions and engaging with the mathematical development.

Taken together, these observations show a consistent pattern across my teaching: strong preparation and mathematical clarity, an approachable and supportive classroom environment, and an increasing emphasis on making student thinking visible during instruction. They have also helped identify a clear direction for continued growth: creating more frequent low-stakes opportunities for students

to predict, discuss, compare approaches, and contribute during class. I have incorporated these recommendations into my continuing development as an instructor through more structured peer discussion, retrieval questions, anonymous polling, and brief opportunities for students to attempt a problem before seeing a complete solution.

Observation Reports:

- [Clare Minnerath — MATH 124: Calculus with Analytic Geometry I](#)
- [Dr. Andrew Loveless — MATH 125: Calculus with Analytic Geometry II](#)
- [Dr. Natalie Naehrig — MATH/STAT 394: Probability I](#)

Individualized instruction

My external tutoring experience has provided a complementary setting for developing teaching skills.

My Superprof profile documents a 5/5 rating from 17 reviews and more than 50 students accompanied through the platform.

Tutoring has repeatedly shown me that a student's visible error is not always the underlying problem. A difficulty with calculus, for example, may originate in algebraic fluency; a difficulty with probability may arise from interpretation rather than computation.

As a result, I have become more deliberate about diagnosing the source of a difficulty before explaining a solution. This habit now influences my classroom teaching: when students produce different answers, I try to determine whether the divergence reflects notation, algebra, conceptual understanding, or interpretation.

Tutoring has also reinforced the importance of gradually reducing instructional support. A successful explanation should not merely allow a student to follow the instructor's reasoning; it should eventually allow the student to reproduce, adapt, and extend that reasoning independently.

Selected Teaching Artifacts

The artifacts below illustrate how my teaching philosophy is implemented in actual course design. Rather than presenting materials without context, I describe the pedagogical purpose of each artifact.

Artifact 1: MATH/STAT 394 course website

Artifact: Public Summer 2026 Probability I website.

[MATH/STAT 394: Probability I](#)

Why this matters. The website illustrates my emphasis on transparency, organization, and accessibility. Students had a central location for the syllabus, homework, solutions, lecture materials, assessment preparation, supplementary resources, and interactive visualizations. Providing both PDF and HTML formats also gave students multiple ways to access course materials.

Artifact 2: Homework 1 and Homework 8 solutions

Artifacts: Sample solutions from the beginning and end of the course.

Why this matters. These materials show how I use written solutions as instructional resources rather than simply answer keys. Including artifacts from both ends of the course also illustrates the progression from foundational probability ideas toward more advanced probabilistic reasoning. The solutions give students a model of mathematical communication that can be revisited independently.

Artifact 3: Midterm examination and preparation materials

Artifacts: Midterm, practice material, review resources, and solutions.

Why this matters. These materials illustrate my approach to transparent assessment. Students should know the concepts and forms of reasoning that an assessment is intended to evaluate. The assessment itself can remain rigorous while review materials reduce unnecessary uncertainty about expectations.

Artifact 4: Final examination

Artifact: Cumulative Probability I final examination.

Why this matters. The final illustrates the cumulative structure of the course and the expectation that students integrate concepts rather than treat each distribution or technique in isolation. It also provides evidence of my experience designing and administering a complete university-level assessment as instructor of record.

Artifact 5: Mini-project

Artifact: Optional Probability I mini-project.

Why this matters. The project created an alternative mode of engagement beyond traditional timed examinations and weekly homework. It reflects my interest in giving students opportunities to work with mathematical ideas in extended and communicative formats.

Mentoring, Advising, and Student Support

Mentoring philosophy

I view mentoring as both academic guidance and translation of the hidden curriculum.

Students often need help not only with mathematical content but also with questions such as: How should I study for this kind of course? How do I know whether I understand a concept? How should I use office hours? How do I recover after a disappointing assessment? How do I turn feedback into a concrete change in study strategy?

My goal is therefore to help students develop systems for learning, not simply solve the immediate problem in front of them.

International student mentoring

At the University of Calgary, I participated in mentoring and community-building activities for international and graduate students. My own experience moving across educational systems made me particularly aware of the academic and social adjustment that can accompany international study.

In mentoring conversations, I emphasized practical navigation of university expectations, connection to available resources, and the normalization of asking for help during transitions.

Individualized tutoring

My work with high-school, undergraduate, and university students has included individualized learning plans, exam preparation, and support for students returning to mathematics after gaps in their education.

These experiences strengthened my ability to adjust the level, pace, and representation of an explanation to the student in front of me rather than relying on a single standard explanation.

Curriculum Development, Educational Leadership, and Service

Curriculum and instructional development

My Summer 2026 Probability I course gave me direct experience with curriculum-level instructional decisions.

I developed the course website, organized the sequence of lecture materials, prepared homework and solutions, designed examinations and review materials, incorporated interactive resources, and revised instructional decisions in response to student needs.

This experience expanded my perspective from the design of an individual quiz section to the alignment of an entire course.

Teaching-related leadership and service

My teaching-related service has included leadership in the Graduate University Mathematics & Statistics Society at the University of Calgary, participation in the Calgary Junior High Math Contest, and community engagement work.

As President of GUMS, I participated in graduate-student advocacy and community-building within mathematics and statistics. This experience reinforced the importance of academic communities in which students have access to peers, information, and opportunities for participation.

My work connected to the Calgary Junior High Math Contest provided experience with mathematical outreach and the evaluation of student mathematical work outside the standard university classroom.

Professional Development in Teaching

I completed the [Graduate Student Certificate in University Teaching and Learning](#) through the Taylor Institute for Teaching and Learning at the University of Calgary.

The program strengthened my understanding of research-informed teaching, course design, assessment, inclusive pedagogy, and reflective practice.

The certificate included the following components.

- **Diversity, Equity, and Inclusion** [[Appendix C.5](#)].

This component focused on equity, diversity, inclusion, and accessibility in postsecondary teaching. It encouraged reflection on barriers to participation, representation, classroom structure, and the design of learning environments that support students from different backgrounds.

- **Emerging Teachers Development** [[Appendix C.2](#)].

This component focused on foundational skills in university teaching, including instructional design, student engagement, classroom facilitation, and reflective professional development.

- **Theories and Practices** [[Appendix C.3](#)].

This component examined theories of student learning, course and curriculum design, assessment, and evidence-based approaches to postsecondary education.

- **Developing Your Teaching Dossier** [[Appendix C.4](#)].

This component focused on articulating a teaching philosophy, connecting that philosophy to instructional practice, and documenting teaching effectiveness through multiple forms of evidence.

Together, these experiences strengthened my ability to connect instructional decisions to pedagogical evidence and to treat teaching as an activity that should be systematically evaluated and refined.

Teaching and Learning Conferences

I value opportunities to engage with broader conversations in higher education, exchange ideas with colleagues, and learn about new approaches to inclusive and evidence-based teaching.

My professional-development activities have included:

MAA MathFest , Boston, Massachusetts	2026
POD Network Conference , Denver, Colorado	2025
Joint Mathematics Meetings (JMM) — MAA Teaching and Pedagogy Sessions , Seattle, Washington	2025
STLHE Annual Conference , Niagara Falls, Ontario	2024
Canadian Society for the Study of Education Annual Conference , Toronto, Ontario	2023
Alberta Mathematics Dialogue , Banff, Alberta	2023

Reflection and Continuous Improvement

I understand teaching as a cycle of planning, implementation, observation, feedback, and revision.

In practice, this means identifying recurring learning bottlenecks, designing targeted supports, and evaluating those supports through student questions, participation, performance patterns, evaluations, and supervisory feedback.

My Summer 2026 Probability I experience provides a useful example.

Several sources of evidence converged on strengths in course organization, clarity, resources, and mathematical explanation. At the same time, student feedback suggested that the transition from classroom examples to difficult homework could sometimes be made more gradual, while the mentor observation suggested increasing the frequency and distribution of student participation.

These findings are complementary rather than contradictory. They suggest that my next stage of development should preserve the strong course infrastructure and mathematical clarity while increasing the amount of intellectual work students do during class.

For a future iteration of Probability I, I would therefore:

- add short “bridge” problems between introductory examples and the most demanding homework problems;
- use more frequent low-stakes prediction and retrieval questions;
- incorporate brief student-to-student discussion before selected whole-class explanations;
- use hybrid tools such as polls and chat more systematically to include remote students;
- move more quickly through routine technical calculations when students already possess the prerequisite skill;
- use the time gained for interpretation, conceptual comparison, and student reasoning.

This process reflects how I understand professional growth in teaching. Effective teaching is not a fixed personal characteristic. It is a scholarly practice that requires evidence, reflection, experimentation, and revision.

Future Teaching Goals and Development Plan

Near-term goals

Over the next year, my primary teaching-development goals are:

1. **Refine independent course design.**

Revise the Probability I materials by strengthening the bridge between lecture examples and the most demanding homework problems, while preserving the conceptual depth of the course.

2. **Increase distributed participation.**

Develop more frequent low-stakes opportunities for every student to predict, retrieve, discuss, or apply mathematical ideas during class.

3. Expand evidence-based assessment.

Continue designing assessments that combine mathematical derivation, conceptual interpretation, method selection, and communication.

4. Develop reusable and accessible course infrastructure.

Continue building public course resources, annotated materials, review guides, interactive demonstrations, and transparent policies that can support future probability, statistics, and mathematics courses.

Longer-term teaching trajectory

Having now served as instructor of record, I aim to deepen my experience with independent instruction and eventually course coordination.

I am particularly interested in teaching probability, statistics, and gateway mathematics courses, where careful course design, transparent expectations, and well-structured instructional support can have substantial effects on students' confidence, persistence, and preparation for later quantitative work.


COURSE SUMMARY REPORT
 Numeric Responses

 University of Washington, Seattle
 College of Arts and Sciences
 Mathematics
 Term: Autumn 2025

 MATH 120 AB
 Precalculus
 Course type: Unknown

Taught by: Arman Jahangiri

Instructor Evaluated: Arman Jahangiri-Grad TA

 Evaluation Delivery: Online
 Evaluation Form: F
 Responses: 7/23 (30% moderate)

Overall Summative Rating represents the combined responses of students to the four global summative items and is presented to provide an overall index of the class's quality:

Combined Median	Adjusted Combined Median
4.2	4.4
(0=lowest; 5=highest)	

Challenge and Engagement Index (CEI) combines student responses to several *IASystem* items relating to how academically challenging students found the course to be and how engaged they were:

CEI: 4.3
(1=lowest; 7=highest)

SUMMATIVE ITEMS

	N	Excellent (5)	Very Good (4)	Good (3)	Fair (2)	Poor (1)	Very Poor (0)	Median	Adjusted Median
The quiz section as a whole was:	7	29%	29%	29%	14%			3.8	4.0
The content of the quiz section was:	7	29%	29%	43%				3.8	4.0
The quiz section instructor's (QSI's) contribution to the course was:	7	57%	29%			14%		4.6	4.8
The QSI's effectiveness in teaching the subject matter was:	7	43%	43%			14%		4.3	4.5

STUDENT ENGAGEMENT

Relative to other college courses you have taken:	N	Much Higher (7)	(6)	(5)	Average (4)	(3)	(2)	Much Lower (1)	Median					
Do you expect your grade in this course to be:	7	14%		29%	43%			14%	4.3					
The intellectual challenge presented was:	7	14%	43%	14%	29%				5.7					
The amount of effort you put into this course was:	7	14%	14%	14%	57%				4.4					
The amount of effort to succeed in this course was:	7	14%	29%		57%				4.4					
Your involvement in course (doing assignments, attending classes, etc.) was:	7	29%		14%	57%				4.4					
On average, how many hours per week have you spent on this course, including attending classes, doing readings, reviewing notes, writing papers and any other course related work?											Class median: 9.0 (N=7)			
Under 2	2-3	4-5	6-7	8-9	10-11	12-13	14-15	16-17	18-19	20-21	22 or more			
	14%	14%		29%		14%	29%							
From the total average hours above, how many do you consider were valuable in advancing your education?											Class median: 7.0 (N=7)			
Under 2	2-3	4-5	6-7	8-9	10-11	12-13	14-15	16-17	18-19	20-21	22 or more			
	14%	14%	29%	14%	14%		14%							
What grade do you expect in this course?											Class median: 3.4 (N=7)			
A (3.9-4.0)	A- (3.5-3.8)	B+ (3.2-3.4)	B (2.9-3.1)	B- (2.5-2.8)	C+ (2.2-2.4)	C (1.9-2.1)	C- (1.5-1.8)	D+ (1.2-1.4)	D (0.9-1.1)	D- (0.7-0.8)	F (0.0)	Pass	Credit	No Credit
14%	29%	29%	14%								14%			
In regard to your academic program, is this course best described as:											(N=7)			
In your major	A core/distribution requirement			An elective		In your minor		A program requirement		Other				
29%	57%							14%						



COURSE SUMMARY REPORT
Numeric Responses

University of Washington, Seattle
College of Arts and Sciences
Mathematics
Term: Autumn 2025

STANDARD FORMATIVE ITEMS

	N	Excellent (5)	Very Good (4)	Good (3)	Fair (2)	Poor (1)	Very Poor (0)	Median	Relative Rank
Explanations by the QSI were:	7	43%	29%	14%		14%		4.2	7
QSI's use of examples and illustrations was:	7	43%	29%	29%				4.2	9
Quality of questions or problems raised by QSI was:	7	57%	29%	14%				4.6	2
QSI's enthusiasm was:	7	43%	43%		14%			4.3	12
Student confidence in QSI's knowledge was:	7	43%	43%		14%			4.3	14
Encouragement given students to express themselves was:	7	71%	14%	14%				4.8	1
Answers to student questions were:	7	57%	29%		14%			4.6	5
Interest level of quiz sections was:	7	29%	29%	29%		14%		3.8	16
QSI's openness to student views was:	7	43%	43%		14%			4.3	11
QSI's ability to deal with student difficulties was:	7	57%	29%			14%		4.6	3
Availability of extra help when needed was:	7	57%	29%	14%				4.6	6
Use of quiz section time was:	7	29%	43%		29%			4.0	13
QSI's interest in whether students learned was:	7	43%	43%		14%			4.3	10
Amount you learned in the quiz sections was:	7	14%	43%	14%	29%			3.7	18
Relevance and usefulness of quiz section content were:	7	43%	43%		14%			4.3	8
Coordination between lectures and quiz sections was:	7	43%	14%	14%	29%			4.0	17
Reasonableness of assigned work for quiz section was:	7	29%	57%		14%			4.1	15
Clarity of student responsibilities and requirements was:	7	57%	29%		14%			4.6	4


COURSE SUMMARY REPORT
 Student Comments

 University of Washington, Seattle
 College of Arts and Sciences
 Mathematics
 Term: Autumn 2025

 MATH 120 AB
 Precalculus
 Course type: Unknown

Taught by: Arman Jahangiri

Instructor Evaluated: Arman Jahangiri-Grad TA

 Evaluation Delivery: Online
 Evaluation Form: F
 Responses: 7/23 (30% moderate)

STANDARD OPEN-ENDED QUESTIONS
Was this class intellectually stimulating? Did it stretch your thinking? Why or why not?

1. This class to me was very well done. The professor was amazing and I mean beyond amazing. He really took the time out of his day and out of his personal time to make sure that we understood exactly what we were learning if you're not doing good or struggling to get by he's very open to helping you.
2. Fair enough
3. Quite hard compared to non college days
4. Arman's advanced understanding of mathematics gave a broader point of view towards the material comparatively to the curriculum itself.

What aspects of this class contributed most to your learning?

1. Definitely the extra review material that he provided along with the Zoom meetings and extra view time for exams
2. Examples
3. Homeworks
4. Arman's willingness to answer questions.

What aspects of this class detracted from your learning?

1. None
2. Enthusiasm
3. Personal lack of effort from me
4. N/A

What suggestions do you have for improving the class?

1. Honestly, I think he's done a very good job at his class. Everything is very well explained I do think in this might be just for me, but I am a visual learner and sometimes I need you to videos or something to help me. Learn since you made so many YouTube videos that was very helpful but to save you some time, maybe see if there's other YouTube videos that explain the same thing just so it's easier on you to not have to make so many and it gives students an opportunity to learn something with maybe a different technique but overall, I think he did amazing
2. Answer questions more clearly or slowly
3. Not sure
4. Less time for warm up - Leave entire class for homework



IASystem Course Summary Reports summarize student ratings of a particular course or combination of courses. They provide a rich perspective on student views by reporting responses in three ways: as frequency distributions, average ratings, and either comparative or adjusted ratings. Remember in interpreting results that it is important to keep in mind the number of students who evaluated the course relative to the total course enrollment as shown on the upper right-hand corner of the report.

Frequency distributions. The percentage of students who selected each response choice is displayed for each item. Percentages are based on the number of students who answered the respective item rather than the number of students who evaluated the course because individual item response is optional.

Median ratings. IASystem reports average ratings in the form of item medians. Although means are a more familiar type of average than medians, they are less accurate in summarizing student ratings. This is because ratings distributions tend to be strongly skewed. That is, most of the ratings are at the high end of the scale and trail off to the low end.

The median indicates the point on the rating scale at which half of the students selected higher ratings, and half selected lower. Medians are computed to one decimal place by interpolation.¹ In general, higher medians reflect more favorable ratings. To interpret median ratings, compare the value of each median to the respective response scale: *Very Poor, Poor, Fair, Good, Very Good, Excellent (0-5); Never/None/Much Lower, About Half/Average, Always/Great/Much Higher (1-7); Slight, Moderate, Considerable, Extensive (1-4).*

Comparative ratings. IASystem provides a normative comparison for each item by reporting the decile rank of the item median. Decile ranks compare the median rating of a particular item to ratings of the same item over the previous two academic years in all classes at the institution and within the college, school, or division. Decile ranks are shown only for items with sufficient normative data.

Decile ranks range from 0 (lowest) to 9 (highest). For all items, higher medians yield higher decile ranks. The 0 decile rank indicates an item median in the lowest 10% of all scores. A decile rank of 1 indicates a median above the bottom 10% and below the top 80%. A decile rank of 9 indicates a median in the top 10% of all scores. Because average ratings tend to be high, a rating of "good" or "average" may have a low decile rank.

Adjusted ratings. Research has shown that student ratings may be somewhat influenced by factors such as class size, expected grade, and reason for enrollment. To correct for this, IASystem reports **adjusted medians** for summative items (items #1-4 and their combined global rating) based on regression analyses of ratings over the previous two academic years in all classes at the respective institution. If large classes at the institution tend to be rated lower than small classes, for example, the adjusted medians for large classes will be slightly higher than their unadjusted medians.

When adjusted ratings are displayed for summative items, **relative rank** is displayed for the more specific (formative) items. Rankings serve as a guide in directing instructional improvement efforts. The top ranked items (1, 2, 3, etc.) represent areas that are going well from a student perspective; whereas the bottom ranked items (18, 17, 16, etc.) represent areas in which the instructor may want to make changes. Relative ranks are computed by first standardizing each item (subtracting the overall institutional average from the item rating for the particular course, then dividing by the standard deviation of the ratings across all courses) and then ranking those standardized scores.

Challenge and Engagement Index (CEI). Several IASystem items ask students how academically challenging they found the course to be. IASystem calculates the average of these items and reports them as a single index. *The Challenge and Engagement Index (CEI)* correlates only modestly with the global rating (median of items 1-4).

Optional items. Student responses to instructor-supplied items are summarized at the end of the evaluation report. Median responses should be interpreted in light of the specific item text and response scale used (response values 1-6 on paper evaluation forms).

¹ For the specific method, see, for example, Guilford, J.P. (1965). *Fundamental statistics in psychology and education*. New York: McGraw-Hill Book Company, pp. 49-53.



COURSE SUMMARY REPORT

Numeric Responses

University of Washington, Seattle
College of Arts and Sciences
Mathematics
Term: Spring 2025

MATH 126 DB
Calculus With Analytic Geometry III
Course type: Face-to-Face
Taught by: Arman Jahangiri
Instructor Evaluated: Arman Jahangiri-Grad TA

Evaluation Delivery: Online
Evaluation Form: F
Responses: 8/39 (21% low)

Overall Summative Rating represents the combined responses of students to the four global summative items and is presented to provide an overall index of the class's quality:

Combined Median	Adjusted Combined Median
4.2	4.3
(0=lowest; 5=highest)	

Challenge and Engagement Index (CEI) combines student responses to several *IASystem* items relating to how academically challenging students found the course to be and how engaged they were:

CEI: 4.9
(1=lowest; 7=highest)

SUMMATIVE ITEMS

	N	Excellent (5)	Very Good (4)	Good (3)	Fair (2)	Poor (1)	Very Poor (0)	Median	Adjusted Median
The quiz section as a whole was:	7	43%		43%	14%			3.3	3.5
The content of the quiz section was:	7	43%	14%	14%	29%			4.0	4.1
The quiz section instructor's (QSI's) contribution to the course was:	7	43%	14%	29%	14%			4.0	4.1
The QSI's effectiveness in teaching the subject matter was:	7	57%	14%	14%	14%			4.6	4.7

STUDENT ENGAGEMENT

Relative to other college courses you have taken:	N	Much Higher (7)	(6)	(5)	Average (4)	(3)	(2)	Much Lower (1)	Median
Do you expect your grade in this course to be:	7		29%	43%	14%	14%			5.0
The intellectual challenge presented was:	7		43%	29%	29%				5.2
The amount of effort you put into this course was:	7	14%	43%	14%	29%				5.7
The amount of effort to succeed in this course was:	7	14%	29%	29%	29%				5.2
Your involvement in course (doing assignments, attending classes, etc.) was:	7	29%	14%	29%	29%				5.2

On average, how many hours per week have you spent on this course, including attending classes, doing readings, reviewing notes, writing papers and any other course related work?

Class median: 7.0 (N=7)

Under 2	2-3	4-5	6-7	8-9	10-11	12-13	14-15	16-17	18-19	20-21	22 or more
14%	14%		29%	14%	14%				14%		

From the total average hours above, how many do you consider were valuable in advancing your education?

Class median: 6.0 (N=7)

Under 2	2-3	4-5	6-7	8-9	10-11	12-13	14-15	16-17	18-19	20-21	22 or more
14%	14%	14%	29%		14%				14%		

What grade do you expect in this course?

Class median: 3.3 (N=6)

A (3.9-4.0)	A- (3.5-3.8)	B+ (3.2-3.4)	B (2.9-3.1)	B- (2.5-2.8)	C+ (2.2-2.4)	C (1.9-2.1)	C- (1.5-1.8)	D+ (1.2-1.4)	D (0.9-1.1)	D- (0.7-0.8)	F (0.0)	Pass	Credit	No Credit
17%	17%	33%	17%			17%								

In regard to your academic program, is this course best described as:

(N=7)

In your major	A core/distribution requirement	An elective	In your minor	A program requirement	Other
43%	29%			29%	



COURSE SUMMARY REPORT
Numeric Responses

University of Washington, Seattle
College of Arts and Sciences
Mathematics
Term: Spring 2025

STANDARD FORMATIVE ITEMS

	N	Excellent (5)	Very Good (4)	Good (3)	Fair (2)	Poor (1)	Very Poor (0)	Median	Relative Rank
Explanations by the QSI were:	7	43%	29%		29%			4.2	5
QSI's use of examples and illustrations was:	7	43%		29%	29%			3.2	15
Quality of questions or problems raised by QSI was:	7	43%		43%	14%			3.3	14
QSI's enthusiasm was:	7	43%		29%	29%			3.2	18
Student confidence in QSI's knowledge was:	7	57%	14%	14%	14%			4.6	4
Encouragement given students to express themselves was:	7	43%		43%	14%			3.3	17
Answers to student questions were:	7	43%	29%	14%	14%			4.2	9
Interest level of quiz sections was:	7	29%	14%	29%	14%	14%		3.2	11
QSI's openness to student views was:	7	57%		29%	14%			4.6	3
QSI's ability to deal with student difficulties was:	7	57%	14%	14%	14%			4.6	1
Availability of extra help when needed was:	7	57%	14%	14%	14%			4.6	2
Use of quiz section time was:	7	29%	29%	14%	29%			3.8	10
QSI's interest in whether students learned was:	7	29%	29%	29%	14%			3.8	12
Amount you learned in the quiz sections was:	7	29%	14%	29%	14%	14%		3.2	13
Relevance and usefulness of quiz section content were:	7	43%		29%	29%			3.2	16
Coordination between lectures and quiz sections was:	7	43%	29%	14%	14%			4.2	6
Reasonableness of assigned work for quiz section was:	7	43%	29%	14%	14%			4.2	7
Clarity of student responsibilities and requirements was:	7	43%	29%	14%	14%			4.2	8



COURSE SUMMARY REPORT

Student Comments

University of Washington, Seattle
College of Arts and Sciences
Mathematics
Term: Spring 2025

MATH 126 DB
Calculus With Analytic Geometry III
Course type: Face-to-Face

Taught by: Arman Jahangiri

Instructor Evaluated: Arman Jahangiri-Grad TA

Evaluation Delivery: Online
Evaluation Form: F
Responses: 8/39 (21% low)

STANDARD OPEN-ENDED QUESTIONS

Was this class intellectually stimulating? Did it stretch your thinking? Why or why not?

1. Doing examples help me understand the concepts
2. Nothing
3. Yes, time spent going over problems was useful and clear
4. Math 126 is amazing. Math but in 3d? Heck yes.

What aspects of this class contributed most to your learning?

1. Thinking skills
2. Nothing
3. Going over practice problems
4. Webassign, EdDiscussion

What aspects of this class detracted from your learning?

1. None
2. Unprepared
3. None
4. Quiz section was somewhat a waste of time. Very helpful if you were falling behind, but didnt push you if you were already ahead.

What suggestions do you have for improving the class?

1. Nothing it's perfect
2. Organization
3. None



IASystem Course Summary Reports summarize student ratings of a particular course or combination of courses. They provide a rich perspective on student views by reporting responses in three ways: as frequency distributions, average ratings, and either comparative or adjusted ratings. Remember in interpreting results that it is important to keep in mind the number of students who evaluated the course relative to the total course enrollment as shown on the upper right-hand corner of the report.

Frequency distributions. The percentage of students who selected each response choice is displayed for each item. Percentages are based on the number of students who answered the respective item rather than the number of students who evaluated the course because individual item response is optional.

Median ratings. IASystem reports average ratings in the form of item medians. Although means are a more familiar type of average than medians, they are less accurate in summarizing student ratings. This is because ratings distributions tend to be strongly skewed. That is, most of the ratings are at the high end of the scale and trail off to the low end.

The median indicates the point on the rating scale at which half of the students selected higher ratings, and half selected lower. Medians are computed to one decimal place by interpolation.¹ In general, higher medians reflect more favorable ratings. To interpret median ratings, compare the value of each median to the respective response scale: *Very Poor, Poor, Fair, Good, Very Good, Excellent (0-5); Never/None/Much Lower, About Half/Average, Always/Great/Much Higher (1-7); Slight, Moderate, Considerable, Extensive (1-4).*

Comparative ratings. IASystem provides a normative comparison for each item by reporting the decile rank of the item median. Decile ranks compare the median rating of a particular item to ratings of the same item over the previous two academic years in all classes at the institution and within the college, school, or division. Decile ranks are shown only for items with sufficient normative data.

Decile ranks range from 0 (lowest) to 9 (highest). For all items, higher medians yield higher decile ranks. The 0 decile rank indicates an item median in the lowest 10% of all scores. A decile rank of 1 indicates a median above the bottom 10% and below the top 80%. A decile rank of 9 indicates a median in the top 10% of all scores. Because average ratings tend to be high, a rating of "good" or "average" may have a low decile rank.

Adjusted ratings. Research has shown that student ratings may be somewhat influenced by factors such as class size, expected grade, and reason for enrollment. To correct for this, IASystem reports **adjusted medians** for summative items (items #1-4 and their combined global rating) based on regression analyses of ratings over the previous two academic years in all classes at the respective institution. If large classes at the institution tend to be rated lower than small classes, for example, the adjusted medians for large classes will be slightly higher than their unadjusted medians.

When adjusted ratings are displayed for summative items, **relative rank** is displayed for the more specific (formative) items. Rankings serve as a guide in directing instructional improvement efforts. The top ranked items (1, 2, 3, etc.) represent areas that are going well from a student perspective; whereas the bottom ranked items (18, 17, 16, etc.) represent areas in which the instructor may want to make changes. Relative ranks are computed by first standardizing each item (subtracting the overall institutional average from the item rating for the particular course, then dividing by the standard deviation of the ratings across all courses) and then ranking those standardized scores.

Challenge and Engagement Index (CEI). Several IASystem items ask students how academically challenging they found the course to be. IASystem calculates the average of these items and reports them as a single index. *The Challenge and Engagement Index (CEI)* correlates only modestly with the global rating (median of items 1-4).

Optional items. Student responses to instructor-supplied items are summarized at the end of the evaluation report. Median responses should be interpreted in light of the specific item text and response scale used (response values 1-6 on paper evaluation forms).

¹ For the specific method, see, for example, Guilford, J.P. (1965). *Fundamental statistics in psychology and education*. New York: McGraw-Hill Book Company, pp. 49-53.



COURSE SUMMARY REPORT

Numeric Responses

University of Washington, Seattle
College of Arts and Sciences
Mathematics
Term: Autumn 2025

MATH 120 AA
Precalculus
Course type: Unknown

Taught by: Arman Jahangiri

Instructor Evaluated: Arman Jahangiri-Grad TA

Evaluation Delivery: Online
Evaluation Form: F
Responses: 14/25 (56% high)

Overall Summative Rating represents the combined responses of students to the four global summative items and is presented to provide an overall index of the class's quality:

Combined Median	Adjusted Combined Median
3.7	3.9
(0=lowest; 5=highest)	

Challenge and Engagement Index (CEI) combines student responses to several *IASystem* items relating to how academically challenging students found the course to be and how engaged they were:

CEI: 4.7
(1=lowest; 7=highest)

SUMMATIVE ITEMS

	N	Excellent (5)	Very Good (4)	Good (3)	Fair (2)	Poor (1)	Very Poor (0)	Median	Adjusted Median
The quiz section as a whole was:	14	29%	14%	50%	7%			3.4	3.6
The content of the quiz section was:	14	21%	36%	36%	7%			3.7	3.9
The quiz section instructor's (QSI's) contribution to the course was:	14	43%	21%	36%				4.2	4.3
The QSI's effectiveness in teaching the subject matter was:	14	43%	7%	43%	7%			3.5	3.7

STUDENT ENGAGEMENT

Relative to other college courses you have taken:	N	Much Higher (7)	(6)	(5)	Average (4)	(3)	(2)	Much Lower (1)	Median
Do you expect your grade in this course to be:	14	7%	21%	14%	50%		7%		4.4
The intellectual challenge presented was:	14	7%	21%	43%	29%				5.0
The amount of effort you put into this course was:	14		36%	36%	21%	7%			5.1
The amount of effort to succeed in this course was:	14	7%	36%	29%	29%				5.2
Your involvement in course (doing assignments, attending classes, etc.) was:	14	29%	14%	29%	29%				5.2

On average, how many hours per week have you spent on this course, including attending classes, doing readings, reviewing notes, writing papers and any other course related work?

Class median: 7.2 (N=13)

Under 2	2-3	4-5	6-7	8-9	10-11	12-13	14-15	16-17	18-19	20-21	22 or more
	8%	15%	31%	15%	8%	15%	8%				

From the total average hours above, how many do you consider were valuable in advancing your education?

Class median: 7.0 (N=13)

Under 2	2-3	4-5	6-7	8-9	10-11	12-13	14-15	16-17	18-19	20-21	22 or more
	15%	23%	15%	23%	8%	15%					

What grade do you expect in this course?

Class median: 3.5 (N=13)

A (3.9-4.0)	A- (3.5-3.8)	B+ (3.2-3.4)	B (2.9-3.1)	B- (2.5-2.8)	C+ (2.2-2.4)	C (1.9-2.1)	C- (1.5-1.8)	D+ (1.2-1.4)	D (0.9-1.1)	D- (0.7-0.8)	F (0.0)	Pass	Credit	No Credit
	69%		15%	8%					8%					

In regard to your academic program, is this course best described as:

(N=13)

In your major	A core/distribution requirement	An elective	In your minor	A program requirement	Other
23%	46%			15%	15%



COURSE SUMMARY REPORT
Numeric Responses

University of Washington, Seattle
College of Arts and Sciences
Mathematics
Term: Autumn 2025

STANDARD FORMATIVE ITEMS

	N	Excellent (5)	Very Good (4)	Good (3)	Fair (2)	Poor (1)	Very Poor (0)	Median	Relative Rank
Explanations by the QSI were:	14	43%	14%	29%	14%			4.0	7
QSI's use of examples and illustrations was:	14	29%	36%	29%	7%			3.9	12
Quality of questions or problems raised by QSI was:	14	36%	21%	36%	7%			3.8	16
QSI's enthusiasm was:	14	29%	36%	29%	7%			3.9	17
Student confidence in QSI's knowledge was:	14	43%	21%	36%				4.2	13
Encouragement given students to express themselves was:	14	36%	36%	21%	7%			4.1	11
Answers to student questions were:	14	43%	14%	36%	7%			4.0	8
Interest level of quiz sections was:	14	29%	21%	43%		7%		3.5	15
QSI's openness to student views was:	14	50%	14%	36%				4.5	4
QSI's ability to deal with student difficulties was:	14	50%	7%	43%				4.5	1
Availability of extra help when needed was:	14	50%	14%	36%				4.5	2
Use of quiz section time was:	14	43%	14%	36%	7%			4.0	6
QSI's interest in whether students learned was:	14	50%	14%	36%				4.5	3
Amount you learned in the quiz sections was:	14	21%	29%	43%	7%			3.5	18
Relevance and usefulness of quiz section content were:	14	36%	29%	29%	7%			4.0	10
Coordination between lectures and quiz sections was:	14	36%	21%	43%				3.8	14
Reasonableness of assigned work for quiz section was:	14	43%	21%	36%				4.2	5
Clarity of student responsibilities and requirements was:	14	29%	43%	29%				4.0	9



COURSE SUMMARY REPORT
Student Comments

University of Washington, Seattle
College of Arts and Sciences
Mathematics
Term: Autumn 2025

MATH 120 AA
Precalculus
Course type: Unknown

Taught by: Arman Jahangiri

Instructor Evaluated: Arman Jahangiri-Grad TA

Evaluation Delivery: Online
Evaluation Form: F
Responses: 14/25 (56% high)

STANDARD OPEN-ENDED QUESTIONS

Was this class intellectually stimulating? Did it stretch your thinking? Why or why not?

1. It did stretch my thinking. Sometimes we have to think about a problem in more than one way.
2. Yes, quiz sections stretched my thinking through applying concepts learned in class to the homework problems.
3. Yes this class intellectually stimulating because this course uses your algebra and geometry skills and you have to really think through the problems and think carefully through them
4. I did find this class intellectually stimulating, I could get my homework done but also get practice on the recent material. When Arman went over the test prep, I understood the material better.
5. yes, it clarified the lecture and helped me make sense of the homework
6. yes, it was challenging.
7. I think this class was intellectually stimulating and stretched my thinking because coming to quiz section was very helpful if I needed help on the homework. The TA was great at answering questions and always attempted to make the most of the short time he was given.
8. I think this quiz section was pretty great, for it being 8:30 in the morning Arman had it figured out and ready even if the questions weren't super clear, he would work his best and the content actually made more sense when he would break it down into different steps to further expand your thinking.
9. To an extent it was. Math is not my best subject so I have learned a lot this year
10. Yes, the application-style problems really made me think deeply about the course content. I think that this course prepared me well for future math classes.

What aspects of this class contributed most to your learning?

1. I liked how the TA would first go over some problems that most people have trouble in.
2. Going through homework problems.
3. Going to office hours and asking for support during quiz sections
4. I really liked how the TA, allowed us to take time to work on the homework. He also went over it. It really helped me out.
5. working out the homework problems in class
6. doing practice problems
7. I think the time given to ask questions about the homework contributed the most to my learning because that was what gave me the most clear understanding on what I was being asked to do.
8. I think the most helpful parts were when class discussions actually led somewhere and were productive to where each problem we talked through led to another equation or new understanding.
9. Being able to work this my classmates
10. The TA was really helpful in answering my questions. He was able to explain complex problems really well. What I learned during quiz section helped me perform better on the exams.

What aspects of this class detracted from your learning?

1. It's an 8:30, and super far away so we are often really tired.
2. Nothing that I can think of.
3. None of the aspects of the class were detracting all of them felt really helpful
4. Nothing
5. N/A
6. n/a
7. I think the test preps that were required at the beginning of most quiz sections is what most detracted from my learning because solving the problems usually took time away from being able to ask questions on the homework problems.
8. I think just the fact that it was a wee bit early made it a little harder to focus because the difference between a 9:30 and 8:30 surprised me but it was very informative.
9. Nothing really.
10. I don't think anything detracted from my learning.

What suggestions do you have for improving the class?

1. Make it not 8:30 and far. Do one or the other, not both.
2. Nothing that I can think of.
3. Allowing retakes on midterms. Doing more reviews on the subjects that we focus on
4. Nothing
5. I would continue to go through the work as a class instead of individual
6. n/a
7. I don't think the test prep should be required. If someone came to class in need of homework help, going through previous exams (if desired by the student) could be done during office hours or on their own time.
8. I don't think I have a lot of suggestions.
9. Nothing. The section wasn't all that useful for me as the way my TA teaches and I learn weren't always the same so it wasn't always that helpful but I got a lot out of being able to help my classmates and I know that my TA helped many of my classmates very well.
10. I would say doing more practice exams during quiz section, which was really helpful at the end of the quarter.



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The median indicates the point on the rating scale at which half of the students selected higher ratings, and half selected lower. Medians are computed to one decimal place by interpolation.¹ In general, higher medians reflect more favorable ratings. To interpret median ratings, compare the value of each median to the respective response scale: *Very Poor, Poor, Fair, Good, Very Good, Excellent (0-5); Never/None/Much Lower, About Half/Average, Always/Great/Much Higher (1-7); Slight, Moderate, Considerable, Extensive (1-4).*

Comparative ratings. IASystem provides a normative comparison for each item by reporting the decile rank of the item median. Decile ranks compare the median rating of a particular item to ratings of the same item over the previous two academic years in all classes at the institution and within the college, school, or division. Decile ranks are shown only for items with sufficient normative data.

Decile ranks range from 0 (lowest) to 9 (highest). For all items, higher medians yield higher decile ranks. The 0 decile rank indicates an item median in the lowest 10% of all scores. A decile rank of 1 indicates a median above the bottom 10% and below the top 80%. A decile rank of 9 indicates a median in the top 10% of all scores. Because average ratings tend to be high, a rating of "good" or "average" may have a low decile rank.

Adjusted ratings. Research has shown that student ratings may be somewhat influenced by factors such as class size, expected grade, and reason for enrollment. To correct for this, IASystem reports **adjusted medians** for summative items (items #1-4 and their combined global rating) based on regression analyses of ratings over the previous two academic years in all classes at the respective institution. If large classes at the institution tend to be rated lower than small classes, for example, the adjusted medians for large classes will be slightly higher than their unadjusted medians.

When adjusted ratings are displayed for summative items, **relative rank** is displayed for the more specific (formative) items. Rankings serve as a guide in directing instructional improvement efforts. The top ranked items (1, 2, 3, etc.) represent areas that are going well from a student perspective; whereas the bottom ranked items (18, 17, 16, etc.) represent areas in which the instructor may want to make changes. Relative ranks are computed by first standardizing each item (subtracting the overall institutional average from the item rating for the particular course, then dividing by the standard deviation of the ratings across all courses) and then ranking those standardized scores.

Challenge and Engagement Index (CEI). Several IASystem items ask students how academically challenging they found the course to be. IASystem calculates the average of these items and reports them as a single index. *The Challenge and Engagement Index (CEI)* correlates only modestly with the global rating (median of items 1-4).

Optional items. Student responses to instructor-supplied items are summarized at the end of the evaluation report. Median responses should be interpreted in light of the specific item text and response scale used (response values 1-6 on paper evaluation forms).

¹ For the specific method, see, for example, Guilford, J.P. (1965). *Fundamental statistics in psychology and education*. New York: McGraw-Hill Book Company, pp. 49-53.



COURSE SUMMARY REPORT

Numeric Responses

University of Washington, Seattle
College of Arts and Sciences
Mathematics
Term: Winter 2026

MATH 125 BB
Calculus With Analytic Geometry II
Course type: Unknown
Taught by: Arman Jahangiri
Instructor Evaluated: Arman Jahangiri-Grad TA

Evaluation Delivery: Online
Evaluation Form: F
Responses: 17/30 (57% high)

Overall Summative Rating represents the combined responses of students to the four global summative items and is presented to provide an overall index of the class's quality:

Combined Median	Adjusted Combined Median
3.7	4.0
(0=lowest; 5=highest)	

Challenge and Engagement Index (CEI) combines student responses to several *IASystem* items relating to how academically challenging students found the course to be and how engaged they were:

CEI: 5.2
(1=lowest; 7=highest)

SUMMATIVE ITEMS

	N	Excellent (5)	Very Good (4)	Good (3)	Fair (2)	Poor (1)	Very Poor (0)	Median	Adjusted Median
The quiz section as a whole was:	17	6%	53%	24%	6%	6%	6%	3.7	4.0
The content of the quiz section was:	17	12%	53%	24%		6%	6%	3.8	4.1
The quiz section instructor's (QSI's) contribution to the course was:	17	24%	35%	18%	12%	6%	6%	3.8	4.0
The QSI's effectiveness in teaching the subject matter was:	17	18%	35%	24%	12%	6%	6%	3.6	3.8

STUDENT ENGAGEMENT

Relative to other college courses you have taken:	N	Much Higher (7)	(6)	(5)	Average (4)	(3)	(2)	Much Lower (1)	Median
Do you expect your grade in this course to be:	17	6%	12%	24%	47%	12%			4.3
The intellectual challenge presented was:	17		53%	35%	6%	6%			5.6
The amount of effort you put into this course was:	17		65%	24%	6%	6%			5.7
The amount of effort to succeed in this course was:	17	6%	53%	24%	12%	6%			5.7
Your involvement in course (doing assignments, attending classes, etc.) was:	17		35%	47%	18%				5.2

On average, how many hours per week have you spent on this course, including attending classes, doing readings, reviewing notes, writing papers and any other course related work?

Class median: 8.0 (N=17)

Under 2	2-3	4-5	6-7	8-9	10-11	12-13	14-15	16-17	18-19	20-21	22 or more
	12%	12%	24%	12%	12%	18%	6%	6%			

From the total average hours above, how many do you consider were valuable in advancing your education?

Class median: 6.0 (N=17)

Under 2	2-3	4-5	6-7	8-9	10-11	12-13	14-15	16-17	18-19	20-21	22 or more
	24%	24%	12%	6%	29%	6%					

What grade do you expect in this course?

Class median: 3.1 (N=17)

A (3.9-4.0)	A- (3.5-3.8)	B+ (3.2-3.4)	B (2.9-3.1)	B- (2.5-2.8)	C+ (2.2-2.4)	C (1.9-2.1)	C- (1.5-1.8)	D+ (1.2-1.4)	D (0.9-1.1)	D- (0.7-0.8)	F (0.0)	Pass	Credit	No Credit
6%	35%	6%	29%	12%		6%						6%		

In regard to your academic program, is this course best described as:

(N=17)

In your major	A core/distribution requirement	An elective	In your minor	A program requirement	Other
29%	29%			41%	



COURSE SUMMARY REPORT
Numeric Responses

University of Washington, Seattle
College of Arts and Sciences
Mathematics
Term: Winter 2026

STANDARD FORMATIVE ITEMS

	N	Excellent (5)	Very Good (4)	Good (3)	Fair (2)	Poor (1)	Very Poor (0)	Median	Relative Rank
Explanations by the QSI were:	17	12%	29%	35%	6%	6%	12%	3.2	16
QSI's use of examples and illustrations was:	17	12%	53%	18%		6%	12%	3.8	10
Quality of questions or problems raised by QSI was:	17	18%	47%	18%	12%		6%	3.8	9
QSI's enthusiasm was:	17	29%	29%	24%	6%		12%	3.8	13
Student confidence in QSI's knowledge was:	17	35%	29%	24%		6%	6%	4.0	12
Encouragement given students to express themselves was:	17	24%	29%	24%	12%		12%	3.6	17
Answers to student questions were:	17	24%	24%	29%	12%		12%	3.4	18
Interest level of quiz sections was:	17	12%	41%	18%	18%		12%	3.6	5
QSI's openness to student views was:	17	35%	35%	18%	6%		6%	4.1	6
QSI's ability to deal with student difficulties was:	16	31%	44%		12%	6%	6%	4.1	1
Availability of extra help when needed was:	17	24%	41%	18%	6%	6%	6%	3.9	11
Use of quiz section time was:	17	18%	47%	18%	6%	6%	6%	3.8	2
QSI's interest in whether students learned was:	17	29%	41%	6%	6%	12%	6%	4.0	7
Amount you learned in the quiz sections was:	17	12%	53%	12%	12%		12%	3.8	3
Relevance and usefulness of quiz section content were:	17	24%	29%	35%			12%	3.6	14
Coordination between lectures and quiz sections was:	17	6%	41%	41%			12%	3.4	15
Reasonableness of assigned work for quiz section was:	17	18%	59%	18%			6%	4.0	4
Clarity of student responsibilities and requirements was:	17	24%	41%	29%			6%	3.9	8



COURSE SUMMARY REPORT

Student Comments

University of Washington, Seattle
College of Arts and Sciences
Mathematics
Term: Winter 2026

MATH 125 BB
Calculus With Analytic Geometry II
Course type: Unknown

Evaluation Delivery: Online
Evaluation Form: F
Responses: 17/30 (57% high)

Taught by: Arman Jahangiri
Instructor Evaluated: Arman Jahangiri-Grad TA

STANDARD OPEN-ENDED QUESTIONS

Was this class intellectually stimulating? Did it stretch your thinking? Why or why not?

1. This class was academically stimulating because it had many questions that were extensions of what was learned in class that were quite unique.
2. Yes. The homework and exams were quite hard but reasonable
3. This class was intellectually stimulating and required me to consider problems beyond the complexity we had learned in class. I had to think through problems not just formulaically but also consider the reasonability of my answers and why they were useful.
4. yes practicing
5. Yes, the application problems helped me understand calculus concepts rather than just memorizing formulas.
6. This class was definitely intellectually stimulating as it required me to not only learn new topics which all connect in various ways, but to apply them to real-life scenarios which stretched my thinking as I tried to visualize the scenarios.
7. Yes, because we had a review material website if we were ever stuck.
8. No, the course posed no challenges as the questions were pretty basic. He also couldn't explain any homework questions and would just give us the answers.
9. Yes, the worksheets were very challenging.
10. It was stimulating and did teach me lots of new techniques.
11. Yes, the math concepts were stimulating especially the word problems, it helped to have different real world instances of the concepts we were learning (motivation)
12. It was intellectually stimulating, but I did not find it stretched my thinking since I had covered most of the content before.
13. This class was definitely intellectually stimulating, it made me expand on ideas I already knew.

What aspects of this class contributed most to your learning?

1. The explanations of problems were certainly very helpful
2. I like that the TA was writing down every problem after we were solving them to explain
3. The practice worksheets were most useful for me in this course. Since they were often previous exam problems, I found them to be highly useful for understanding the difficulty and structure of questions we would actually be asked about.
4. test prep
5. quiz section, TA going over problems after we had time to solve them
6. The explanations in lecture and general patterns/formulas explained to us for different kinds of problems contribute most to my learning because they truly helped "separate my confusions" before they took root, which would usually create mental blocks and discourage me from attempting to solve similar problems.
7. Loveless' review materials.
8. Nothing.
9. the worksheets
10. The problem walk-throughs were very helpful in my learning.
11. The worksheet sections were good for practicing what we learned in class.
12. group worksheets
13. Probably the homework and previous exams.
14. The board explanations were really useful, always learning the same thing but in a different way helps to understand it better!

What aspects of this class detracted from your learning?

2. None
3. The worksheets in the longer quiz section felt less connected to the actual course content and weren't as useful when I reviewed for exams.
4. n/a
5. worksheets in quiz section (Tuesday ones were helpful, but the ones we did on Thursday were not as relevant to the type of problems that appear on the test)
6. When the TA and professor provided differing methods of solving concepts so that I felt like I had just failed to understand the concept as a whole since I clearly couldn't follow the new explanation.

- 7. N/A
- 8. The lack of help available.
- 9. the test prep
- 10. Some of the topics were a little disjointed.
- 11. I didn't like how far test prep section was and sometimes it was hard to do the test prep because I missed lecture (but it was still good for doing my homework)
- 12. instructor went over the answers to examples too fast, I would lose track of what was going on or it would make me lose my train of thought with my own work.
- 13. The quiz section.

What suggestions do you have for improving the class?

- 2. I don't have any suggestions
- 3. I'd suggest having all the time in the quiz section spent on previous exam problems, since they were incredibly useful in preparing me for the exams.
- 4. i would love to have note summarized from TA like what should i put on the cheat sheet for the exam. i know dr loveless has a lot of notes but im not sure if i can put them all on my cheat sheet
- 5. Have a few quizzes so there is less weighting on 2 midterms and 1 test
- 6. It would be helpful for the TA to be completely aware of the professor's explanations so as o avoid confusing students who are being introduced to the material for the first time. Also, when explaining problems and concepts to the class, the TA should not assume it to be a review for the students but rather teach and explain each step to at least the first few problems. This method greatly benefits students who may have been confused on topics but who, without a detailed step-by-step explanation of the approaches, would not even know what questions to ask.
- 7. Perhaps more stressing of practice before the first midterm the same way we were highly encouraged to do practice before the second midterm.
- 8. Make it optional and theres no need to do worksheets or test preps.
- 9. no
- 10. None. The teacher was good at giving clear explanations.
- 11. Maybe have people work on the test prep in groups like in quiz section
- 12. do things more as a class instead of our own, I liked the small groups and when we worked together but it was hard because we weren't encouraged to collaborate.
- 13. Nothing



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Informal TA Observation Report

Instructor: Arman

Date Observed: February 10, 2026

Course: Math 125 Section (7.2)

Observer: Dr. Andrew Loveless

Arman began the section by distributing a handout and giving students time (about five minutes) to begin working. He circulated around the room, checked several students' work, stopped to talk with individuals, and made it around the entire room. Students appeared engaged and were actively working and chatting about the problems. The focus of the discussion included trigonometric integrals (e.g., integrals involving secant and tangent). Arman later transitioned to going over prepared notes at the board.

Strengths Observed*1. Strong physical presence and approachability*

Arman did an excellent job circulating early. He moved around the entire classroom, checked student work, and spoke individually with multiple students. This created an active working environment and signaled approachability. I especially liked how you wandered and chatted with students — that lowers anxiety and makes section feel collaborative rather than evaluative.

2. Students were engaged and working diligently

There was productive noise in the room. Students were comparing work and discussing ideas. That is a very positive sign in a calculus section. It reflects both comfort and intellectual effort.

3. Clear mathematical knowledge and organization

When reviewing problems at the board, Arman demonstrated strong content knowledge. He spoke clearly, explained steps logically, and was organized in his presentation. The mathematical flow was easy to follow.

4. Calm and steady classroom tone

The atmosphere felt low-pressure and steady. Students seemed comfortable working and talking. This tone is particularly important in Math 125, where many students can feel anxious about technique-heavy topics.

5. Well prepared

You were well prepared with detailed solutions. That preparation shows and contributes to a smooth class experience. Students benefit from seeing problems laid out clearly and coherently.

Teaching Tip: Homework Check-In at the Start

One idea that could add even more responsiveness to your section would be a “check-in” at the start:

- Put up homework due dates. Ask if there are specific homework problems students are stuck on.
- Have them call out numbers (e.g., 8.1 #4, 9.1 #10) and write them on the board.

Then, while you circulate, you can glance at those problems and decide whether to build time around one of them. You can also ask directly: “Do you want to see one of these worked out?” or “What was your question about on that homework problem?” This creates ownership and signals that section is responsive to their needs. It also helps you target your board time to what students actually want.

Suggestions to Try

These are refinements — the foundation is strong.

1. *Increase structured student-to-student engagement*

Students were already talking during work time. You might intentionally amplify that by prompting interaction:

- “Check with your neighbor.”
- “Did anyone do this a different way?”
- “Raise your hand if you factored it differently.”
- “Turn and compare Step 1.”

If a few students solved something differently, briefly ask them what they did. Those small back-and-forth build mathematical conversation. You can also occasionally use visible commitment checks:

- “Raise your hand if you feel good about this.”
- “Thumbs up / sideways / down.”

These short moments often significantly increase engagement.

2. *Use wandering time to find common errors that you bring back to the group*

While you circulate (which you do well), consider identifying 2–3 common errors that multiple students are making. Then mention them explicitly: “An error I saw a few times while walking around was this...” This helps everyone watch for those pitfalls and reinforces that you are responding to their actual thinking.

3. *Clarify technique emphasis*

For the integral $\int \sec^4(x) \tan^3(x) dx$, you correctly noted that it can be handled by pulling out either $\sec^2(x)$ or $\sec(x)\tan(x)$, and both substitution paths work well. You also demonstrated integration by parts. While that can technically work, I would recommend skipping it here. It is harder for students to generalize and less obvious how to choose parts. In this section, focusing on one broadly applicable technique will likely reduce confusion and build confidence.

4. *Board visibility*

I would suggest writing larger and spreading your work out more (or enlarging content on the screen if using prepared notes). That makes it easier to point to specific parts and ask:

- “What comes next?”
- “What identity are we using?”
- “What should we label here?”

You are somewhat constrained by the room setup, but small adjustments in spacing and size can make interaction easier.

Overall Impression

This was a strong and productive section. Students were working seriously and engaging with the math. Your early circulation and calm presence are clear strengths. The next step be increasing intentional whole-class interaction. You are building a solid classroom presence. Thank you for your thoughtful work with our students.

Arman Jahangiri Observation

Observer: Clare Minnerath

October 15, 2024

I observed Arman as their TA mentor during Arman's first quarter teaching at the University of Washington. Arman was leading a worksheet section on limits and derivatives. The observation began about 25 minutes into a 120 minute section where there were about 25 students in attendance. Arman was working through worksheet questions at the board with class participation.

Strengths

1. *Organization and Clarity*

The graph from the worksheet was very neatly drawn on the board, Arman's voice was clear, and at the end of each part Arman asked if there were questions. When students did have questions Arman repeated them and clarified that they heard the question correctly.

2. *Confidence and Calm Demeanor*

At one point Arman made a mistake in their work at the board. When a student pointed it out, Arman admitted to the mistake, clarified to the class why they made it, and thanked the student for pointing it out.

3. *Approachability*

A student gave an incomplete answer. Arman validated the student's partial answer as correct and guided the class to the complete answer.

Suggestions

1. Incorporating more group work or student discussion time.
2. Eliciting more student participation by asking leading questions or "Think, Pair, Share" exercises.

Overall, Arman appeared confident and experienced at the board. Arman was eager to implement the suggestions, and we continued discussion on how to best make the changes without isolating students less comfortable participating.

Informal TA Observation Report

Instructor: Arman

Date Observed: February 10, 2026

Course: Math 125 Section (7.2)

Observer: Dr. Andrew Loveless

Arman began the section by distributing a handout and giving students time (about five minutes) to begin working. He circulated around the room, checked several students' work, stopped to talk with individuals, and made it around the entire room. Students appeared engaged and were actively working and chatting about the problems. The focus of the discussion included trigonometric integrals (e.g., integrals involving secant and tangent). Arman later transitioned to going over prepared notes at the board.

Strengths Observed

1. Strong physical presence and approachability

Arman did an excellent job circulating early. He moved around the entire classroom, checked student work, and spoke individually with multiple students. This created an active working environment and signaled approachability. I especially liked how you wandered and chatted with students — that lowers anxiety and makes section feel collaborative rather than evaluative.

2. Students were engaged and working diligently

There was productive noise in the room. Students were comparing work and discussing ideas. That is a very positive sign in a calculus section. It reflects both comfort and intellectual effort.

3. Clear mathematical knowledge and organization

When reviewing problems at the board, Arman demonstrated strong content knowledge. He spoke clearly, explained steps logically, and was organized in his presentation. The mathematical flow was easy to follow.

4. Calm and steady classroom tone

The atmosphere felt low-pressure and steady. Students seemed comfortable working and talking. This tone is particularly important in Math 125, where many students can feel anxious about technique-heavy topics.

5. Well prepared

You were well prepared with detailed solutions. That preparation shows and contributes to a smooth class experience. Students benefit from seeing problems laid out clearly and coherently.

Teaching Tip: Homework Check-In at the Start

One idea that could add even more responsiveness to your section would be a “check-in” at the start:

- Put up homework due dates. Ask if there are specific homework problems students are stuck on.
- Have them call out numbers (e.g., 8.1 #4, 9.1 #10) and write them on the board.

Then, while you circulate, you can glance at those problems and decide whether to build time around one of them. You can also ask directly: “Do you want to see one of these worked out?” or “What was your question about on that homework problem?” This creates ownership and signals that section is responsive to their needs. It also helps you target your board time to what students actually want.

Suggestions to Try

These are refinements — the foundation is strong.

1. *Increase structured student-to-student engagement*

Students were already talking during work time. You might intentionally amplify that by prompting interaction:

- “Check with your neighbor.”
- “Did anyone do this a different way?”
- “Raise your hand if you factored it differently.”
- “Turn and compare Step 1.”

If a few students solved something differently, briefly ask them what they did. Those small back-and-forth build mathematical conversation. You can also occasionally use visible commitment checks:

- “Raise your hand if you feel good about this.”
- “Thumbs up / sideways / down.”

These short moments often significantly increase engagement.

2. *Use wandering time to find common errors that you bring back to the group*

While you circulate (which you do well), consider identifying 2–3 common errors that multiple students are making. Then mention them explicitly: “An error I saw a few times while walking around was this...” This helps everyone watch for those pitfalls and reinforces that you are responding to their actual thinking.

3. *Clarify technique emphasis*

For the integral $\int \sec^4(x) \tan^3(x) dx$, you correctly noted that it can be handled by pulling out either $\sec^2(x)$ or $\sec(x)\tan(x)$, and both substitution paths work well. You also demonstrated integration by parts. While that can technically work, I would recommend skipping it here. It is harder for students to generalize and less obvious how to choose parts. In this section, focusing on one broadly applicable technique will likely reduce confusion and build confidence.

4. *Board visibility*

I would suggest writing larger and spreading your work out more (or enlarging content on the screen if using prepared notes). That makes it easier to point to specific parts and ask:

- “What comes next?”
- “What identity are we using?”
- “What should we label here?”

You are somewhat constrained by the room setup, but small adjustments in spacing and size can make interaction easier.

Overall Impression

This was a strong and productive section. Students were working seriously and engaging with the math. Your early circulation and calm presence are clear strengths. The next step be increasing intentional whole-class interaction. You are building a solid classroom presence. Thank you for your thoughtful work with our students.



**GRADUATE STUDENT
CERTIFICATE OF COMPLETION IN
UNIVERSITY TEACHING AND LEARNING**

This certificate is presented to

Arman Jahangiri

Offered by the Taylor Institute for Teaching and Learning

A handwritten signature in black ink, appearing to read "Cheryl Jeffs".

Cheryl Jeffs, EdD
Certificates in University Teaching and
Learning, Graduate Student Lead

A handwritten signature in black ink, appearing to read "Wendy Benoit".

Wendy Benoit, PhD
Vice-Provost (Teaching and Learning)

2023-24



Arman Jahangiri

has successfully completed the requirements for

Emerging Teachers Development: Graduate Students

Taylor Institute for Teaching and Learning

October 21, 2025



Criteria:

<https://badges.ucalgary.ca/badges/66>

Verification:

<https://badges.ucalgary.ca/achievements/b6cf4514a5b89e8ca40269d8cc59acfa>





Arman Jahangiri

has successfully completed the requirements for

Theories and Practices: Graduate Students

Taylor Institute for Teaching and Learning

October 21, 2025



Criteria:

<https://badges.ucalgary.ca/badges/72>

Verification:

<https://badges.ucalgary.ca/achievements/f63c7a3c28c511878188da89a632c64b>





Arman Jahangiri

has successfully completed the requirements for

Developing Your Teaching Dossier: Graduate Students

Taylor Institute for Teaching and Learning

October 21, 2025



Criteria:

<https://badges.ucalgary.ca/badges/64>

Verification:

<https://badges.ucalgary.ca/achievements/b987843d772de139e722195947a263af>





Arman Jahangiri

has successfully completed the requirements for

EDI Badge for Graduate Students

Taylor Institute for Teaching and Learning

October 21, 2025



Criteria:

<https://badges.ucalgary.ca/badges/220>

Verification:

<https://badges.ucalgary.ca/achievements/dd5a34e55ccf98c1dd9475fd22d0c91b>





COURSE SUMMARY REPORT Numeric Responses

University of Washington, Seattle
College of Arts and Sciences
Mathematics
Term: Summer 2026

MATH 394 B, Joint with STAT 394 B
Probability I
Course type: Unknown

Evaluation Delivery: Online
Evaluation Form: A
Responses: 14/21 (67% high)

Taught by: Arman Jahangiri

Instructor Evaluated: Arman Jahangiri-Predoc Inst

Overall Summative Rating represents the combined responses of students to the four global summative items and is presented to provide an overall index of the class's quality:

Combined Median	Adjusted Combined Median
4.1	4.1
(0=lowest; 5=highest)	

Challenge and Engagement Index (CEI) combines student responses to several *IASystem* items relating to how academically challenging students found the course to be and how engaged they were:

CEI: 5.0
(1=lowest; 7=highest)

SUMMATIVE ITEMS

	N	Excellent (5)	Very Good (4)	Good (3)	Fair (2)	Poor (1)	Very Poor (0)	Median	Adjusted Median
The course as a whole was:	14	29%	36%	21%	14%			3.9	3.9
The course content was:	14	29%	43%	29%				4.0	4.0
The instructor's contribution to the course was:	14	50%	29%	14%	7%			4.5	4.5
The instructor's effectiveness in teaching the subject matter was:	14	29%	43%	14%	7%	7%		4.0	4.0

STUDENT ENGAGEMENT

Relative to other college courses you have taken:	N	Much Higher (7)	(6)	(5)	Average (4)	(3)	(2)	Much Lower (1)	Median	
Do you expect your grade in this course to be:	13	8%	31%	15%	38%	8%			4.8	
The intellectual challenge presented was:	13	8%	38%	31%	15%	8%			5.4	
The amount of effort you put into this course was:	13	15%	31%	31%	23%				5.4	
The amount of effort to succeed in this course was:	13	23%	23%	31%	23%				5.4	
Your involvement in course (doing assignments, attending classes, etc.) was:	13	15%	23%	23%	38%				5.0	

On average, how many hours per week have you spent on this course, including attending classes, doing readings, reviewing notes, writing papers and any other course related work?

Class median: 10.0 Hours per credit: 2.5 (N=11)

Under 2	2-3	4-5	6-7	8-9	10-11	12-13	14-15	16-17	18-19	20-21	22 or more
		9%	9%	27%	18%	18%	9%	9%			

From the total average hours above, how many do you consider were valuable in advancing your education?

Class median: 8.5 Hours per credit: 2.1 (N=11)

Under 2	2-3	4-5	6-7	8-9	10-11	12-13	14-15	16-17	18-19	20-21	22 or more
9%	18%	9%	9%	9%	27%	9%	9%				

What grade do you expect in this course?

Class median: 3.5 (N=11)

A (3.9-4.0)	A- (3.5-3.8)	B+ (3.2-3.4)	B (2.9-3.1)	B- (2.5-2.8)	C+ (2.2-2.4)	C (1.9-2.1)	C- (1.5-1.8)	D+ (1.2-1.4)	D (0.9-1.1)	D- (0.7-0.8)	F (0.0)	Pass	Credit	No Credit
36%	18%	18%	9%	9%	9%									

In regard to your academic program, is this course best described as:

(N=12)

In your major	A core/distribution requirement	An elective	In your minor	A program requirement	Other
33%	8%	17%	17%	17%	8%



COURSE SUMMARY REPORT
Numeric Responses

University of Washington, Seattle
College of Arts and Sciences
Mathematics
Term: Summer 2026

STANDARD FORMATIVE ITEMS

	N	Excellent (5)	Very Good (4)	Good (3)	Fair (2)	Poor (1)	Very Poor (0)	Median	Relative Rank
Course organization was:	13	23%	38%	15%	23%			3.8	15
Clarity of instructor's voice was:	13	54%	23%	23%				4.6	8
Explanations by instructor were:	13	38%	38%	8%	8%	8%		4.2	10
Instructor's ability to present alternative explanations when needed was:	12	33%	25%	33%	8%			3.8	18
Instructor's use of examples and illustrations was:	13	62%	23%	8%	8%			4.7	1
Quality of questions or problems raised by the instructor was:	12	33%	33%	17%	17%			4.0	16
Student confidence in instructor's knowledge was:	12	50%	33%	17%				4.5	9
Instructor's enthusiasm was:	12	42%	42%		17%			4.3	14
Encouragement given students to express themselves was:	11	55%	36%			9%		4.6	6
Answers to student questions were:	12	50%	25%	8%	17%			4.5	5
Availability of extra help when needed was:	12	50%	33%	8%	8%			4.5	7
Use of class time was:	12	33%	42%	17%		8%		4.1	12
Instructor's interest in whether students learned was:	12	42%	50%		8%			4.3	11
Amount you learned in the course was:	12	33%	25%	8%	25%	8%		3.8	17
Relevance and usefulness of course content were:	12	50%	17%	17%	17%			4.5	3
Evaluative and grading techniques (tests, papers, projects, etc.) were:	12	58%	33%	8%				4.6	2
Reasonableness of assigned work was:	12	42%	25%	25%	8%			4.2	13
Clarity of student responsibilities and requirements was:	12	50%	25%	17%	8%			4.5	4



COURSE SUMMARY REPORT
Student Comments

University of Washington, Seattle
College of Arts and Sciences
Mathematics
Term: Summer 2026

MATH 394 B, Joint with STAT 394 B
Probability I
Course type: Unknown

Evaluation Delivery: Online
Evaluation Form: A
Responses: 14/21 (67% high)

Taught by: Arman Jahangiri

Instructor Evaluated: Arman Jahangiri-Predoc Inst

STANDARD OPEN-ENDED QUESTIONS

Was this class intellectually stimulating? Did it stretch your thinking? Why or why not?

1. It was. The content itself touches many branches of math and connects them, which was my favorite part of this course.
2. Yes, this class was intellectually stimulating. It helped me think more carefully about probability and solve problems in different ways.
3. Yes, as an economics major this class has been extremely helpful. I was able to enhance my understanding in the content I learned in my economics class by learning more mathematical sides to it.
4. Class taught me a lot of subject of math i'm not very familiar with
7. I found this class very interesting!
8. absolutely. even for a math nerd there were some good concepts

What aspects of this class contributed most to your learning?

1. The homework assignments really helped to cement my understanding of the material.
2. The examples, practice problems, and clear explanations contributed the most to my learning.
3. I was able to learn about different distributions (in economics I had only learned normal distribution) and a deeper analysis regarding statistics.
4. The course website was very well made and organized and all the materials prepared for exam prep were very useful.
5. I liked having the slides to look back on, and the updated slides with instructor notes. Having homework solutions posted and recorded lectures was so good for looking back on information and learning from my mistakes, I wish all math classes did that.
6. Having slides posted and annotated slides posted, and having zoom recordings available to review after lectures whenever needed
7. I liked that Arman made visualizations of some of the concepts we were learning. Seeing a visual representation of the concepts helped a lot with my understanding.
8. good homework problems.

What aspects of this class detracted from your learning?

1. I felt that every so often there would be a homework question that was unreasonably hard and take hours upon hours to solve.
2. Sometimes the class moved a little too quickly, so some difficult topics were hard to understand.
3. N/A
4. Me being very unfamiliar with a lot of the notation used in the class hurt my learning experience. It also felt there was a substantial gap between the hw and content taught in class.
5. The way the slides were organized sometimes felt confusing. If I got lost in lecture, it wasn't always clear exactly what concept we were working on. Might have been my issue for getting lost in lecture. Sometimes the homework assignments did not feel related enough to what we covered in lecture. I think more examples might help with this. Sometimes I googled questions because I did not feel like I had a strong enough understanding of how a concept applied to a situation.
7. I sometimes the lectures and the homeworks were pretty different. More examples that connect to the homework in later lectures would've been better.
8. none

What suggestions do you have for improving the class?

1. Other than some of the homework problems being too hard and time consuming, I thought this was an excellent class with little to improve on.
2. I suggest spending more time on difficult topics and providing more practice problems before exams.
3. Keep this up. He did an excellent job in explaining the concepts and creating homeworks and exams.
4. More examples related to the hw in class would be nice
5. The homework assignments for a certain week's content should be due the week after we learn about it. Even though that is a big break from the day the content is taught to the day the homework is due, I feel it is worse to be working on Sunday's homework assignment on Wednesday and feeling like I don't know enough to get even half of it done.
8. keep leaning into those homework problems and the proofs, i really appreciated them.



IASystem Course Summary Reports summarize student ratings of a particular course or combination of courses. They provide a rich perspective on student views by reporting responses in three ways: as frequency distributions, average ratings, and either comparative or adjusted ratings. Remember in interpreting results that it is important to keep in mind the number of students who evaluated the course relative to the total course enrollment as shown on the upper right-hand corner of the report.

Frequency distributions. The percentage of students who selected each response choice is displayed for each item. Percentages are based on the number of students who answered the respective item rather than the number of students who evaluated the course because individual item response is optional.

Median ratings. IASystem reports average ratings in the form of item medians. Although means are a more familiar type of average than medians, they are less accurate in summarizing student ratings. This is because ratings distributions tend to be strongly skewed. That is, most of the ratings are at the high end of the scale and trail off to the low end.

The median indicates the point on the rating scale at which half of the students selected higher ratings, and half selected lower. Medians are computed to one decimal place by interpolation.¹ In general, higher medians reflect more favorable ratings. To interpret median ratings, compare the value of each median to the respective response scale: *Very Poor, Poor, Fair, Good, Very Good, Excellent (0-5); Never/None/Much Lower, About Half/Average, Always/Great/Much Higher (1-7); Slight, Moderate, Considerable, Extensive (1-4).*

Comparative ratings. IASystem provides a normative comparison for each item by reporting the decile rank of the item median. Decile ranks compare the median rating of a particular item to ratings of the same item over the previous two academic years in all classes at the institution and within the college, school, or division. Decile ranks are shown only for items with sufficient normative data.

Decile ranks range from 0 (lowest) to 9 (highest). For all items, higher medians yield higher decile ranks. The 0 decile rank indicates an item median in the lowest 10% of all scores. A decile rank of 1 indicates a median above the bottom 10% and below the top 80%. A decile rank of 9 indicates a median in the top 10% of all scores. Because average ratings tend to be high, a rating of "good" or "average" may have a low decile rank.

Adjusted ratings. Research has shown that student ratings may be somewhat influenced by factors such as class size, expected grade, and reason for enrollment. To correct for this, IASystem reports **adjusted medians** for summative items (items #1-4 and their combined global rating) based on regression analyses of ratings over the previous two academic years in all classes at the respective institution. If large classes at the institution tend to be rated lower than small classes, for example, the adjusted medians for large classes will be slightly higher than their unadjusted medians.

When adjusted ratings are displayed for summative items, **relative rank** is displayed for the more specific (formative) items. Rankings serve as a guide in directing instructional improvement efforts. The top ranked items (1, 2, 3, etc.) represent areas that are going well from a student perspective; whereas the bottom ranked items (18, 17, 16, etc.) represent areas in which the instructor may want to make changes. Relative ranks are computed by first standardizing each item (subtracting the overall institutional average from the item rating for the particular course, then dividing by the standard deviation of the ratings across all courses) and then ranking those standardized scores.

Challenge and Engagement Index (CEI). Several IASystem items ask students how academically challenging they found the course to be. IASystem calculates the average of these items and reports them as a single index. *The Challenge and Engagement Index (CEI)* correlates only modestly with the global rating (median of items 1-4).

Optional Items. Student responses to instructor-supplied items are summarized at the end of the evaluation report. Median responses should be interpreted in light of the specific item text and response scale used (response values 1-6 on paper evaluation forms).

¹ For the specific method, see, for example, Guilford, J.P. (1965). *Fundamental statistics in psychology and education*. New York: McGraw-Hill Book Company, pp. 49-53.

Teaching Observation Report

Arman Jahangiri – Math 397: Probability I

Exponential Distribution, Expected Value, Variance, and Memorylessness

TA observed	Arman Jahangiri
Course	Math 397: Probability I
Session type	Hybrid lecture
Topic	Exponential distribution, expected value, variance, memorylessness, and related probability concepts
Date	July 24, 2026

Overall Impression

This was an effective and carefully prepared hybrid lecture. Arman provided students with unusually strong organizational support through a well-structured course website, clear review materials, posted solutions, and neatly designed lecture notes. His explanations were calm, well paced, and well matched to the course level. Students received a coherent introduction to the exponential distribution that also reinforced earlier course ideas, including probability density functions and the geometric distribution.

The main opportunity for growth is not the mathematical clarity of the session, which was consistently strong, but the level and distribution of student participation. In a small class of about seven in-person students and several students on Zoom, there is considerable room to build a more interactive learning community.

Strengths and Pedagogical Impact

Arman began by situating the class within the broader course. His comments on the midterm, posted solutions, review sheet, and mini-project component communicated that the course has a clear structure and that students are being supported in multiple ways. The review sheet, in particular, gave students a useful way to identify the main ideas of the course and prepare purposefully. Making the midterm closely aligned with the practice and review materials also created a transparent assessment environment: students could focus their preparation on understanding the intended learning goals rather than guessing what might appear.

The course materials were a major strength. The organized GitHub course page and exceptionally neat notes reduce unnecessary organizational burden for students and make it easier to revisit ideas after class. Arman's summary slide at the end of the section and his use of learning goals when beginning new topics further helped students identify what mattered most.

Arman also made effective conceptual connections. When introducing the exponential distribution, he invited students to recall the geometric distribution and used the comparison to help distinguish the two. Similarly, he used new material as an opportunity to review how to verify that a function is a probability density function. These connections support durable learning because students are encouraged to organize probability concepts into a network of related ideas rather than treating each distribution as a separate procedure.

The Poll Everywhere question was another particularly effective choice. After working through one example, Arman gave students three minutes to apply a similar idea independently and reminded them to ask questions if they became stuck. The split between correct and incorrect responses made the debrief pedagogically valuable: instead of simply supplying the answer, Arman asked students to consider whether errors arose from algebra or from conceptual understanding. This kind of check gives both students and instructor useful information about what needs attention.

Arman's explanations of why expected value, variance, and medians matter were also valuable. Brief motivation through data matching and connections to later courses helped students see these quantities as tools with purposes, rather than formulas to memorize. During the proof of the memoryless property, he invited input on an idea students should already know. This was a good move because it asked students to retrieve and apply prior knowledge within a new argument.

Student Engagement and Learning Environment

The classroom atmosphere was calm, supportive, and professionally organized. Students asked thoughtful questions, including questions that extended from Arman's comments, which suggests genuine interest in the material. Students also appeared to follow parts of the memorylessness proof closely, as reflected in visible nodding.

At the same time, the overall participation pattern was fairly quiet. Much of the class appeared to be copying notes, and one student seemed to answer a disproportionate number of questions. A few students were also attending to phones or screens rather than the class at one point. These observations do not outweigh the strengths of the session, but they suggest that students would benefit from more regular invitations to process and contribute during the lecture.

Arman already has a strong starting point in the Poll Everywhere activity and in the questions he asks during proofs. Extending these practices could make student thinking more visible and distribute participation more widely. In such a small class, learning and using students' names would be especially helpful. Addressing students personally can strengthen connection, communicate that each student's contribution matters, and make it easier to invite a broader range of voices into the conversation.

Areas for Continued Growth

A useful next step would be to create more frequent, low-stakes moments for students to respond, predict, or briefly work before an explanation is completed. For example, during a proof, Arman might ask students to identify the next step, compare two distributions with a neighbor or through Zoom chat, or decide which earlier result is most relevant. These moments need not be lengthy. Their value is that they require students to retrieve and use ideas while the lecture is happening, rather than only record a finished solution.

A second refinement concerns pacing at familiar technical steps. Arman's careful detail was generally helpful, but in a Math 397 course some calculations, such as the mechanics of integration by parts in an expected-value computation, may not require the same amount of in-class attention. Keeping the full derivation available in the notes while moving more efficiently through standard algebraic steps could create time for a conceptual question, a brief student activity, or a discussion of interpretation.

Protocol Alignment

1. Classroom Organization & Clarity of Purpose

Protocol points addressed:

- Clear structure and pacing.
- Students understand the goal or purpose of the session.
- Logical flow of explanations or activities.

Evidence and interpretation. The session was organized around a clear progression of review, new material, examples, and a concluding summary. Learning goals and summary slides clarified the intended focus, while the lecture connected exponential distributions to earlier concepts and moved from explanation to individual application and debrief.

2. Communication & Instructional Delivery

Protocol points addressed:

- Clear explanations and mathematical communication.
- Effective use of board/visuals.
- Delivery students can follow, including pace, visibility, and audibility.

Evidence and interpretation. Arman's explanations were calm, appropriately paced, and well matched to the course level. The carefully organized notes, review sheet, course website, and summary slide gave students strong visual and organizational support. Comparisons, motivation, and explicit review helped students follow both new concepts and their relation to earlier material.

3. Student Engagement & Participation

Protocol points addressed:

- Students actively think, respond, or work.
- TA encourages participation and gives wait time.
- TA listens to and builds on student ideas.

Evidence and interpretation. Poll Everywhere gave students a meaningful opportunity to apply a similar problem independently, and students contributed answers and questions. Arman invited input during the lecture and asked students to diagnose the source of errors after the poll. More frequent low-stakes prompts could make engagement more visible and reduce reliance on a small number of frequent respondents.

4. Learning Environment & Rapport

Protocol points addressed:

- Welcoming, respectful tone.
- Encouragement of student questions.
- Constructive responses to confusion or errors.
- Approachability.
- Evidence of student focus, including any periods of distraction and possible causes.

Evidence and interpretation. The session had a calm, supportive tone, and Arman's course materials and assessment transparency communicated care for student success. Students asked questions beyond the immediate examples, indicating that questions were welcome. Students appeared attentive during important parts of the proof, though the overall room was quiet and a few students were off task. Learning and using names could strengthen connection and help invite participation from every student in this small hybrid class.

5. Support for Student Learning

Protocol points addressed:

- Checks for understanding.
- Addresses common difficulties.
- Connects the session to broader course goals.
- Students leave knowing the key idea of the day.

Evidence and interpretation. The Poll Everywhere question gave Arman clear evidence of students' understanding and misunderstanding. Asking whether errors were algebraic or conceptual was a productive way to identify the kind of support students needed. Arman connected distributions to one another and motivated expected value, variance, and medians through their uses and future relevance. The learning goals, repeated conceptual connections, and final summary supported students in identifying the central takeaways.

6. Group Work Organization: Not Applicable

Formal group work was not part of the observed session.

Summary

Arman gave a strong, well-prepared hybrid lecture. His clear explanations, excellent course materials, purposeful review, and thoughtful use of Poll Everywhere supported students' conceptual understanding of the exponential distribution and its connections to earlier probability topics. The principal refinement is to build more frequent opportunities for students to contribute and to use the small class size to create a more personal, participatory learning community.

MATH/STAT 394: Probability I

University of Washington

Syllabus

MATH/STAT 394: Probability I

Summer 2026

University of Washington

Instructor:	Arman Jahangiri armanjg@uw.edu
Time & Location:	MWF, 12:00–1:00, THO 125
Modality:	Online Learning – Hybrid
Official Course Website:	https://canvas.uw.edu/courses/1896341
Instructor’s Course Website:	https://armanjg.github.io/MATH-394-Probability-1/
Office Hours:	Mondays 1:00-2:00 and Thursdays 2:00-3:00 at PDL 109 and Zoom

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Welcome

Welcome to MATH/STAT 394: Probability I! Probability is one of the central languages of modern mathematics, statistics, data science, machine learning, and decision-making under uncertainty. The course is designed to develop both technical fluency and probabilistic intuition.

This syllabus is subject to reasonable adjustment during the quarter. Any changes will be announced on Canvas.

Course Description

MATH/STAT 394 is the first course in the introductory probability sequence (MATH/STAT 394-395). Students will learn how to understand and translate uncertain situations into probability models, perform exact and approximate calculations, and interpret results clearly.

Topics generally include axiomatic definitions of probability; random variables; conditional probability and Bayes' theorem; expectations and variance; named distributions: binomial, geometric, Poisson, uniform (discrete and continuous), normal and exponential; normal and Poisson approximations to binomial. Transformations of a single random variable. Markov and Chebyshev's inequality. Weak law of large numbers for finite variance.

Prerequisites

The formal prerequisite is one of MATH 126 or MATH 136.

Students should be comfortable with reading and writing mathematical explanations, some set theory, limits, continuity, differentiation/integration techniques, and sums/series.

Learning Objectives

By the end of the course, students will be able to:

1. define and use sample spaces, events, probability measures, random variables, distributions, conditional probability, independence, expectation, and variance;
2. formulate an uncertainty problem as a probability model;
3. compute probabilistic quantities such as expectations, variances, and distribution functions for discrete and continuous random variables;
4. recognize and use well-known distributions such as Bernoulli, binomial, geometric, Poisson, uniform, exponential, normal, etc.;
5. find the distribution of random variables and functions of random variables;
6. explain the meaning and use of the WLLN (Weak Law of Large Numbers), the CLT (Central Limit Theorem), and their applications such as Monte Carlo integration, approximating and bounding probabilistic quantities.

Resources, Textbooks and References

Main Resources

- Lecture Slides (links will be made available a day before each class on instructor's website)
- Joe Blitzstein and Jessica Hwang, *Introduction to Probability*:
Free online version:
<https://drive.google.com/file/d/1VmkAAGOYCTORq1wxSQqy255qLJjTNvBI/edit>

Optional Resources

- Hossein Pishro-Nik, *Introduction to Probability, Statistics, and Random Processes*:
<https://www.probabilitycourse.com>
- Harvard STAT 110 course website by Joe Blitzstein:
<https://stat110.hsites.harvard.edu/>
- Charles M. Grinstead and J. Laurie Snell, *Introduction to Probability*:
<https://math.dartmouth.edu/~prob/prob/prob.pdf>
- David F. Anderson, Timo Seppäläinen, and Benedek Valkó, *Introduction to Probability*, Cambridge University Press.
- Sheldon Ross, *A First Course in Probability*.
- William Feller, *An Introduction to Probability Theory and Its Applications*, Vol. I.
- Kai Lai Chung, *Elementary Probability Theory with Stochastic Processes*.
- Hoel, Port, and Stone, *Introduction to Probability Theory*.

Communication

- **Canvas**: Announcements, assignments, grades, and schedule updates will be posted there.
- **Instructor Website**: Lecture slides, course documents, homework assignments and solutions, and additional course resources will be posted there.
- **Ed Discussion**: Students are encouraged to post conceptual, homework, and logistics questions there, since other students may benefit from the answers. Anonymous posting may be used when appropriate.
- **Gradescope**: Homework submission, exam grading, and regrade requests will be handled there.
- **Instructor Email**: The instructor may be contacted for private, personal, or urgent matters.

Please allow up to 48 hours for an email response during the work week. Emails sent late Friday or during the weekend may receive a response the following week.

Grading

Final grades will be determined tentatively according Tables 1 and 2.

The following grading benchmarks are approximate; exact grade cutoffs may vary slightly and are not guaranteed. Grades between the listed benchmarks will be determined by linear interpolation. Exceptions to the above table may be considered in certain situations.

Table 1: Course Components and Weights

Component	Weight
Homework	40%
Midterm	25%
Final	35%
Bonus*	5%

Table 2: Approximate Grading Benchmarks

Percent Earned	Course GPA
95	4.0
90	3.7
85	3.4
80	3.1
75	2.8
70	2.5
65	2.2
60	2.0
55	1.7
50	1.3
45	1.0
40	0.7

(★) **Attendance (3%)** Students with fewer than 4 absences throughout the quarter will receive the full attendance score (3%). Beginning with the 4th absence, each additional absence will reduce the attendance score by 1

(★) **Mini-Project (2%)** Students will have the opportunity to earn an additional 2% bonus by recording a short presentation teaching a probabilistic concept not covered in the course. Additional alternative opportunities for earning this bonus may also be announced throughout the quarter.

Exam Policies

Examinations are closed-book and closed-notes. Students may use:

- one handwritten 8.5×11 inch reference sheet (front and back), and
- a non-programmable non-graphing calculator capable of standard exponential and logarithmic computations (e.g. TI-30X IIS).

No other electronic devices, calculators, communication devices, or external resources may be used during examinations.

All **regrade requests** must be submitted through Gradescope, within **three** days of the grade being released.

Homework Policy

Submission and Deadline

- Homework must be submitted as a single PDF on Gradescope by **10:00 PM (Pacific Time)** on the listed due date.
- Each problem should begin on a new page unless otherwise stated.
- Pages must be tagged correctly in Gradescope.
- Work must be written clearly and legibly. Illegible work may receive little or no credit.
- LaTeX is strongly encouraged but not required.

Late Days

Each student is allotted **six late days** for the quarter. A late day extends a homework deadline by up to 24 hours without penalty. For example, submitting an assignment anytime between 10:01 PM on the due date and 10:00 PM the following day counts as one late day.

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Once all late days have been exhausted, additional late submissions will incur a penalty of **10% per day**, up to a maximum deduction of **50%**. Assignments submitted more than five days late, or after solutions have been released, will not be accepted.

Submission Issues and Technical Difficulties

If a serious technical issue prevents a timely Gradescope submission, students may temporarily submit their assignment by email to the instructor (armanjg@uw.edu). In such cases:

- The submission must consist of a **single PDF file**.
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HW# Submission - FULL NAME - STUDENT ID

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- The student should clearly indicate in the Gradescope submission that the assignment was previously submitted by email before the deadline.

Email submissions are intended only for genuine technical emergencies and should not be used as a substitute for timely Gradescope submission.

Collaboration Policy

Students are encouraged to discuss homework ideas with classmates. However, submitted work must be written independently and in each student's own words.

If collaboration occurs, collaborators' names should be listed at the top of the submission. Copying solutions from another student, from the internet, from solution manuals, or from AI tools without genuine understanding and an original write-up is not accepted.

A useful guideline is to discuss ideas, then close all outside sources and write the solution independently.

Canvas, Ed Discussion, Gradescope, Zoom, and possibly R/RMarkdown or L^AT_EX may be used throughout the course.

Students are expected to have regular access to a reliable internet connection and a device capable of accessing course materials and completing assignments.

The University of Washington also provides technology-related support resources for students, including equipment programs through services such as the Student Technology Loan Program (STLP) and other UW technology initiatives. These programs may provide temporary access to laptops, webcams, hotspots, and other educational technology resources for eligible students.

Accessibility and Disability Accommodations

Student access and inclusion are important priorities in this course. The University of Washington is committed to creating inclusive and accessible learning environments consistent with federal and state law.

Students who have already established accommodations with Disability Resources for Students (DRS) should activate their accommodations through myDRS as soon as possible so that implementation can be arranged. Students who have not established accommodations but have a temporary health condition or permanent disability that may require accommodation should contact DRS directly.

DRS website: <https://disability.uw.edu>

Religious Accommodations

Washington state law requires that the University of Washington develop a policy for accommodation of student absences or significant hardship due to reasons of faith or conscience, or for organized religious activities. The UW's policy, including additional information about how to request an accommodation, is available at [Religious Accommodations Policy](#).

Accommodations must be requested within the first two weeks of the course using the [Religious Accommodations Request Form](#).

Diversity, Inclusion, and Classroom Climate

Diverse backgrounds, experiences, and perspectives are essential to university learning. Everyone in this course is expected to treat others with respect. This includes respect across differences of age, disability, ethnicity, family status, gender identity and expression, immigration status, national origin, race, religion, political belief, sex, sexual orientation, socioeconomic status, veteran status, and other identities or experiences.

Harassment, discrimination, intimidation, or disrespectful behavior is not acceptable in class, office hours, online discussions, or course communications.

Academic Integrity

Academic integrity is essential to the learning community. Academic misconduct may include submitting work that was not understood or independently produced, using unauthorized materials during exams, sharing exam content or solutions, posting or obtaining course solutions from unauthorized websites, and representing someone else's work as one's own.

Use of AI Tools

AI tools can sometimes be useful for studying, checking notation, generating practice questions, or debugging LaTeX. However, they can also produce incorrect solutions and can significantly interfere with learning when used as a substitute for independent thinking. AI-generated solutions may not be submitted as original work. Any permitted use of AI must support, not replace, the student's own reasoning.

Tips for Success

- Attend the lectures actively.
- Start homework early.
- Check Canvas and use Ed Discussion regularly.
- Attempt problems seriously before seeking help.
- Attend office hours.
- Work with classmates.

Safety and Health

Prioritizing your well-being throughout the quarter is essential. Try to support both your physical and mental health by maintaining balanced habits. Limiting harmful coping mechanisms, including substance use, can also make a significant difference in your overall resilience. Building healthy routines can improve your ability to manage pressure, stay focused on your objectives, and navigate difficult periods more effectively. Remember that challenging times affect many people, and seeking guidance or encouragement from others is important. Reaching out for help early often prevents stress from becoming overwhelming.

MATH/STAT 394: Probability I

University of Washington

Tentative Weekly Schedule

This schedule is approximate and may change.

Week	Lecture	Date	Topic
1	1	Jun 22 (M)	Chapter 1A — Sample Spaces and Events
1	2	Jun 24 (W)	Chapter 1B — Naive Probability
1	3	Jun 26 (F)	Chapter 1C–1D — Counting
		Jun 28 (S)	HW 1 due
2	4	Jun 29 (M)	Chapter 1C–1D — Counting
2	5	Jul 01 (W)	Chapter 1F — Axiomatic Probability
2	–	Jul 03 (F)	University Holiday — Independence Day Observed (no class)
		Jul 05 (S)	HW 2 due
3	6	Jul 06 (M)	Chapter 1F — Axiomatic Probability
3	7	Jul 08 (W)	Chapter 2 — Conditional Probability
3	8	Jul 10 (F)	Chapter 2 — Conditional Probability
		Jul 12 (S)	HW 3 due
4	9	Jul 13 (M)	Chapter 3 — Random Variables
4	10	Jul 15 (W)	Chapter 3 — Random Variables
4	11	Jul 17 (F)	Chapter 4 — Expectation
		Jul 19 (S)	HW 4 due
5	12	Jul 20 (M)	Chapter 4 — Expectation
5	13	Jul 22 (W)	Midterm Exam
5	14	Jul 24 (F)	Chapter 4 — Expectation / Chapter 5A — Continuous Uniform
		Jul 26 (S)	HW 5 due
6	15	Jul 27 (M)	Chapter 5B — Normal and Exponential
6	16	Jul 29 (W)	Chapter 6A — Moments
6	17	Jul 31 (F)	Chapter 6B — MGFs
		Aug 02 (S)	HW 6 due
7	18	Aug 03 (M)	Chapter 7 — Joint Distributions
7	19	Aug 05 (W)	Chapter 7 — Joint Distributions
7	20	Aug 07 (F)	Chapter 8A — Transformations
		Aug 09 (S)	HW 7 due
8	21	Aug 10 (M)	Chapter 8B — Beta and Gamma
8	22	Aug 12 (W)	Chapter 10A — Inequalities
8	23	Aug 14 (F)	Chapter 10B — Law of Large Numbers and Central Limit Theorem
		Aug 16 (S)	HW 8 due
9	24	Aug 17 (M)	Chapter 10C — Chi-Square and Student t -Distributions
9	25	Aug 19 (W)	Review for Final
9	26	Aug 21 (F)	Final Exam

University of Washington
Department of Mathematics

Homework 1 Solution

MATH 394: Probability I

Instructor: Arman Jahangiri

Submission: Single PDF on Gradescope
Deadline: 10:00 PM Pacific Time on the listed due date

This homework is worth **50 points**.

Across the quarter, there are **eight homework assignments**, worth **400 points total**. Homework assignments together account for **40%** of the final course grade. Further course policies can be seen in the [MATH 394 syllabus](#).

MATH 394: Probability I**Homework****Homework Policy****Submission and Deadline**

Please:

- Submit homework as a single PDF on Gradescope by **10:00 PM (Pacific Time)** on the listed due date.
- Begin each problem on a new page, and ensure that pages are correctly tagged in Gradescope.
- Write solutions clearly and legibly.
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MATH 394: Probability I

Homework

Problem 1.

Source: Blitzstein and Hwang, *Introduction to Probability*, Chapter 1, Problem 23.

Three people get into an empty elevator at the first floor of a building that has 10 floors. Each presses the button for their desired floor, unless one of the others has already pressed that button. Assume that they are equally likely to want to go to floors 2 through 10, independently of each other.

What is the probability that the buttons for 3 consecutive floors are pressed?

Solution

There are 9 possible destination floors, namely floors 2 through 10. Since the three people make their choices independently, it is natural to count ordered triples of desired floors. The total number of such triples is

$$9^3.$$

For exactly three consecutive buttons to be pressed, the three desired floors must be distinct, and the set of floors must look like

$$\{j, j+1, j+2\}, \quad \text{with } j = 2, \dots, 8.$$

The smallest possible starting floor is 2 and the largest possible starting floor is 8, so there are 7 possible consecutive triples of floors. Once such a triple of floors has been chosen, the three people may be assigned to those floors in $3!$ different orders. Thus the number of favorable ordered triples is

$$7 \cdot 3!.$$

Hence the desired probability is

$$\frac{7 \cdot 3!}{9^3} = \frac{42}{729} = \frac{14}{243}.$$

MATH 394: Probability I

Homework

Problem 2.

Source: Blitzstein and Hwang, *Introduction to Probability*, Chapter 1, Problem 3.

Fred is planning to go out to dinner each night of a certain week, Monday through Friday, with each dinner being at one of his ten favorite restaurants.

- (a) How many possibilities are there for Fred's schedule of dinners for that Monday through Friday, if Fred is not willing to eat at the same restaurant more than once?
- (b) How many possibilities are there for Fred's schedule of dinners for that Monday through Friday, if Fred is willing to eat at the same restaurant more than once, but is not willing to eat at the same place twice in a row or more?

Solution

(a) Fred is choosing restaurants for five ordered nights, and in this part no restaurant may be repeated. There are 10 choices for Monday, then 9 choices for Tuesday, then 8 choices for Wednesday, then 7 choices for Thursday, and finally 6 choices for Friday. Therefore the number of possible dinner schedules is

$$10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 = 30240.$$

(b) Now Fred may return to a restaurant later in the week, but he may not eat at the same restaurant on two consecutive nights. Monday can be chosen in 10 ways. On each of the remaining four nights, exactly one restaurant is forbidden, namely the restaurant from the previous night. Thus each of Tuesday through Friday has 9 available choices. The number of possible schedules is

$$10 \cdot 9^4 = 65610.$$

MATH 394: Probability I

Homework

Problem 3.

Source: Blitzstein and Hwang, *Introduction to Probability*, Chapter 1, Problem 15–18.

Give story proofs for the following identities.

(a) Show that

$$\sum_{k=0}^n \binom{n}{k} = 2^n.$$

(b) Show that

$$\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n},$$

for all positive integers n .

(c) Show that

$$\sum_{k=1}^n k \binom{n}{k}^2 = n \binom{2n-1}{n-1},$$

for all positive integers n .

Hint: Consider choosing a committee of size n from two groups of size n each, where only one of the two groups has people eligible to become the chair of the committee.

Solution

(a) Consider a group of n people, and imagine forming a committee from this group. One way to make the committee is to decide, separately for each person, whether that person is included. Since each person has two possibilities, included or not included, the number of committees is 2^n .

The same committees can also be counted by their size. If the committee has k members, then it can be chosen in $\binom{n}{k}$ ways. Since the committee may have any size from 0 to n , the total number of committees is

$$\sum_{k=0}^n \binom{n}{k}.$$

These two expressions count the same set of committees, and so

$$\sum_{k=0}^n \binom{n}{k} = 2^n.$$

(b) Suppose there are two groups, each containing n people. We want to form a committee of size n from the combined group of $2n$ people. If we ignore the two-group structure, this can be done in

$$\binom{2n}{n}$$

ways.

Now count the same committees by the number k chosen from the first group. If k people are chosen from the first group, then $n - k$ people must be chosen from the second group. This gives

$$\binom{n}{k} \binom{n}{n-k}$$

MATH 394: Probability I

Homework

committees. Since $\binom{n}{n-k} = \binom{n}{k}$, the number with this value of k is $\binom{n}{k}^2$. Summing over all possible values of k gives

$$\sum_{k=0}^n \binom{n}{k}^2.$$

Both counts describe the same committees, and hence

(c) Again take two groups of n people each. We form a committee of size n , and then choose a chair from the first group among those who are on the committee.

Count first according to the number k of committee members who come from the first group. The first group contributes k people in $\binom{n}{k}$ ways, and the second group contributes the remaining $n - k$ people in $\binom{n}{n-k} = \binom{n}{k}$ ways. Once the committee has been chosen, there are k choices for the chair, since the chair must be one of the selected members from the first group. Thus this count gives

$$\sum_{k=1}^n k \binom{n}{k}^2.$$

There is a shorter count of the same chaired committees. Choose the chair first from the first group, which gives n choices. After the chair is chosen, we still need $n - 1$ additional committee members, and they may be chosen from the remaining $2n - 1$ people. This gives

$$n \binom{2n-1}{n-1}.$$

The two counts are counting exactly the same objects, so

MATH 394: Probability I

Homework

Problem 4.

Source: Blitzstein and Hwang, *Introduction to Probability*, Chapter 1, Problem 25.

A city with 6 districts has 6 robberies in a particular week. Assume the robberies are located randomly, with all possibilities for which robbery occurred where equally likely.

What is the probability that some district had more than 1 robbery?

Solution

It is easiest to count the complement. Treat the 6 robberies as distinguishable, for instance by the order in which they occurred. Each robbery may occur in any of the 6 districts, so there are

$$6^6$$

equally likely assignments.

The complement of the event in the problem is the event that no district had more than one robbery. Since there are 6 robberies and 6 districts, this means that each district had exactly one robbery. The number of assignments with exactly one robbery in each district is $6!$, since the robberies must be matched one-to-one with the districts.

Therefore

$$\mathbb{P}(\text{no district had more than one robbery}) = \frac{6!}{6^6}.$$

It follows that

$$\mathbb{P}(\text{some district had more than one robbery}) = 1 - \frac{6!}{6^6} = 1 - \frac{720}{46656} = \frac{319}{324}.$$

MATH 394: Probability I

Homework

Problem 5.

Source: Blitzstein and Hwang, *Introduction to Probability*, Chapter 1, Problem 5.

A knock-out tournament is being held with 2^n tennis players. This means that for each round, the winners move on to the next round and the losers are eliminated, until only one person remains. For example, if initially there are $2^4 = 16$ players, then there are 8 games in the first round, then the 8 winners move on to round 2, then the 4 winners move on to round 3, then the 2 winners move on to round 4, the winner of which is declared the winner of the tournament. There are various systems for determining who plays whom within a round, but these do not matter for this problem.

- (a) How many rounds are there?
- (b) Count how many games in total are played, by adding up the numbers of games played in each round.
- (c) Count how many games in total are played, this time by directly thinking about it without doing almost any calculation.

Hint: How many players need to be eliminated?

Solution

(a) There are 2^n players at the beginning. After each round, half of the players remain. Thus the numbers of players remaining after successive rounds are

$$2^n, 2^{n-1}, 2^{n-2}, \dots, 2, 1.$$

It takes n rounds to reduce the field from 2^n players to one champion.

(b) In the first round, 2^{n-1} games are played. In the second round, 2^{n-2} games are played. Continuing in this way, the final round has one game. The total number of games is therefore

$$2^{n-1} + 2^{n-2} + \dots + 2 + 1 = 2^n - 1.$$

(c) There is also a very direct way to see the same number. Every game eliminates exactly one player. To finish with one champion, the tournament must eliminate all the other players. Since there are 2^n players at the start, the number of eliminated players is

$$2^n - 1.$$

Thus the number of games is $2^n - 1$.

MATH 394: Probability I

Homework

Problem 6.

Source: Blitzstein and Hwang, *Introduction to Probability*, Chapter 1, Problem 49.

A fair die is rolled n times. What is the probability that at least 1 of the 6 values never appears?

Solution

There are 6^n equally likely sequences of n die rolls. We want the probability that at least one of the six faces is absent.

Let A_j be the event that face j never appears. The desired event is

$$A_1 \cup A_2 \cup \cdots \cup A_6.$$

We count this union by inclusion-exclusion. If a particular set of k faces is missing, then each roll must be one of the remaining $6 - k$ faces, so there are

$$(6 - k)^n$$

sequences. There are $\binom{6}{k}$ ways to choose the missing faces. Hence, using the inclusion-exclusion principle, we have

$$\mathbb{P}(A_1 \cup \cdots \cup A_6) = \frac{1}{6^n} \sum_{k=1}^6 (-1)^{k+1} \binom{6}{k} (6 - k)^n.$$

Writing the expression out gives

MATH 394: Probability I

Homework

Problem 7.

Source: Blitzstein and Hwang, *Introduction to Probability*, Chapter 1, Problem 53.

Fred needs to choose a password for a certain website. Assume that he will choose an 8-character password, and that the legal characters are the lowercase letters $a, b, c, \dots, z$, the uppercase letters $A, B, C, \dots, Z$, and the numbers $0, 1, \dots, 9$.

- How many possibilities are there if he is required to have at least one lowercase letter in his password?
- How many possibilities are there if he is required to have at least one lowercase letter and at least one uppercase letter in his password?
- How many possibilities are there if he is required to have at least one lowercase letter, at least one uppercase letter, and at least one number in his password?

Solution

There are 62 legal characters in total, namely 26 lowercase letters, 26 uppercase letters, and 10 digits.

(a) Count all 8-character passwords and subtract those with no lowercase letters. There are 62^8 passwords in total. If no lowercase letter appears, each character must be chosen from the 26 uppercase letters and the 10 digits, giving 36^8 such passwords. Therefore the number with at least one lowercase letter is

Final Answer for Part (a)

$$62^8 - 36^8$$

(b) Now the password must contain at least one lowercase letter and at least one uppercase letter. Start with all 62^8 passwords. Subtract the passwords with no lowercase letters and the passwords with no uppercase letters. Each of these two forbidden classes has 36^8 passwords. Passwords with neither lowercase nor uppercase letters use only digits, and there are 10^8 of them. These digit-only passwords have been subtracted twice, so they must be added back once. The desired number is

Final Answer for Part (b)

$$62^8 - 2 \cdot 36^8 + 10^8$$

(c) Finally, require all three types of characters to appear. Again begin with 62^8 total passwords. Subtract the passwords missing lowercase letters, missing uppercase letters, or missing digits. These counts are 36^8 , 36^8 , and 52^8 , respectively.

Now add back the passwords in which two of the three character types are missing. If lowercase and uppercase letters are both missing, only digits remain, giving 10^8 passwords. If lowercase letters and digits are both missing, only uppercase letters remain, giving 26^8 passwords. If uppercase letters and digits are both missing, only lowercase letters remain, again giving 26^8 passwords. Since a password cannot be missing all three character types and still have length 8, there is no further term. Thus the number of valid passwords is

MATH 394: Probability I**Homework****Final Answer for Part (c)**

$$62^8 - 2 \cdot 36^8 - 52^8 + 10^8 + 2 \cdot 26^8$$

University of Washington
Department of Mathematics

Homework 8 Solutions

MATH 394: Probability I

Instructor: Arman Jahangiri

Submission: Single PDF on Gradescope
Deadline: 10:00 PM Pacific Time on the listed due date

This homework is worth **50 points**.

Across the quarter, there are **eight homework assignments**, worth **400 points total**. Homework assignments together account for **40%** of the final course grade. Further course policies can be seen in the [MATH 394 syllabus](#).

MATH 394: Probability I**Homework 8 Solutions****Homework Policy****Submission and Deadline**

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MATH 394: Probability I

Homework 8 Solutions

Problem 1.

Source: Blitzstein and Hwang, *Introduction to Probability*, Chapter 8, Problem 1–5.

Find the following PDFs.

- (a) Find the PDF of e^{-X} for $X \sim \text{Exp}(1)$.
- (b) Find the PDF of X^7 for $X \sim \text{Exp}(\lambda)$.
- (c) Find the PDF of Z^3 for $Z \sim N(0, 1)$.
- (d) Find the PDF of Z^4 for $Z \sim N(0, 1)$.
- (e) Find the PDF of $|Z|$ for $Z \sim N(0, 1)$.

Solution

Let $\phi(z) = (2\pi)^{-1/2}e^{-z^2/2}$ denote the standard normal density.

(a) Put $Y = e^{-X}$. Since $X > 0$, we have $0 < Y < 1$. The inverse transformation is $x = -\log y$, and

$$\left| \frac{dx}{dy} \right| = \frac{1}{y}.$$

Thus

$$f_Y(y) = e^{-(-\log y)} \frac{1}{y} = 1, \quad 0 < y < 1.$$

(b) Put $Y = X^7$. The inverse is $x = y^{1/7}$ for $y > 0$, and

$$\left| \frac{dx}{dy} \right| = \frac{1}{7}y^{-6/7}.$$

Therefore

$$f_Y(y) = \frac{\lambda}{7}y^{-6/7}e^{-\lambda y^{1/7}}, \quad y > 0.$$

(c) Put $Y = Z^3$. This transformation is one-to-one. For $y \neq 0$, the inverse is $z = y^{1/3}$ and

$$\left| \frac{dz}{dy} \right| = \frac{1}{3|y|^{2/3}}.$$

Hence

$$f_Y(y) = \frac{\phi(y^{1/3})}{3|y|^{2/3}}, \quad y \neq 0.$$

The value assigned to the density at the single point $y = 0$ is immaterial.

(d) Put $Y = Z^4$. The support is $y > 0$. For each $y > 0$, the two preimages are $y^{1/4}$ and $-y^{1/4}$. Adding the two contributions gives

$$f_Y(y) = \phi(y^{1/4}) \frac{1}{4y^{3/4}} + \phi(-y^{1/4}) \frac{1}{4y^{3/4}} = \frac{\phi(y^{1/4})}{2y^{3/4}}, \quad y > 0.$$

(e) Put $Y = |Z|$. For $y > 0$, the two preimages are y and $-y$, and therefore

$$f_Y(y) = \phi(y) + \phi(-y) = 2\phi(y), \quad y > 0.$$

MATH 394: Probability I

Homework 8 Solutions

Problem 2.

Source: Blitzstein and Hwang, *Introduction to Probability*, Chapter 8, Problem 13.

Let X and Y be i.i.d. $\text{Exp}(\lambda)$, and let

$$T = \log(X/Y).$$

Find the CDF and PDF of T .

Solution

For real t ,

$$F_T(t) = P(\log(X/Y) \leq t) = P(X \leq e^t Y).$$

Conditioning on $Y = y$ gives

$$F_T(t) = \int_0^\infty P(X \leq e^t y) \lambda e^{-\lambda y} dy.$$

Since $X \sim \text{Exp}(\lambda)$,

$$P(X \leq e^t y) = 1 - e^{-\lambda e^t y}.$$

Thus

$$F_T(t) = \int_0^\infty (1 - e^{-\lambda e^t y}) \lambda e^{-\lambda y} dy = 1 - \frac{1}{1 + e^t} = \frac{e^t}{1 + e^t}.$$

Differentiating the CDF gives

$$f_T(t) = \frac{e^t}{(1 + e^t)^2}, \quad -\infty < t < \infty.$$

The parameter λ disappears because only the ratio X/Y matters.

MATH 394: Probability I

Homework 8 Solutions

Problem 3.

Source: Blitzstein and Hwang, *Introduction to Probability*, Chapter 8, Problem 14.

Let X and Y have joint PDF $f_{X,Y}(x,y)$, and transform $(X,Y) \mapsto (T,W)$ linearly by letting

$$T = aX + bY, \quad W = cX + dY,$$

where a, b, c, d are constants such that

$$ad - bc \neq 0.$$

- (a) Find the joint PDF $f_{T,W}(t,w)$ in terms of $f_{X,Y}$, though your answer should be written as a function of t and w .

- (b) For the case where

$$T = X + Y, \quad W = X - Y,$$

show that

$$f_{T,W}(t,w) = \frac{1}{2} f_{X,Y} \left(\frac{t+w}{2}, \frac{t-w}{2} \right).$$

Solution

- (a) Let

$$D = ad - bc.$$

Solving the linear system gives

$$x = \frac{dt - bw}{D}, \quad y = \frac{-ct + aw}{D}.$$

The Jacobian determinant of the inverse transformation is

$$\left| \frac{\partial(x,y)}{\partial(t,w)} \right| = \frac{1}{|D|}.$$

Therefore

$$f_{T,W}(t,w) = \frac{1}{|ad - bc|} f_{X,Y} \left(\frac{dt - bw}{ad - bc}, \frac{-ct + aw}{ad - bc} \right),$$

for all (t,w) whose inverse point lies in the support of (X,Y) .

- (b) Here $a = 1$, $b = 1$, $c = 1$, and $d = -1$, so

$$ad - bc = -2.$$

Also

$$x = \frac{t+w}{2}, \quad y = \frac{t-w}{2}.$$

Thus

$$f_{T,W}(t,w) = \frac{1}{2} f_{X,Y} \left(\frac{t+w}{2}, \frac{t-w}{2} \right).$$

MATH 394: Probability I

Homework 8 Solutions

Problem 4.

Source: Blitzstein and Hwang, *Introduction to Probability*, Chapter 10, Problem 2.

For i.i.d. random variables $X_1, \dots, X_n$ with mean μ and variance σ^2 , give a value of n , as a specific number, that will ensure that there is at least a 99% chance that the sample mean will be within 2 standard deviations of the true mean μ .

Solution

The sample mean has

$$E(\bar{X}_n) = \mu, \quad \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

Chebyshev's inequality gives

$$P(|\bar{X}_n - \mu| \geq 2\sigma) \leq \frac{\text{Var}(\bar{X}_n)}{(2\sigma)^2} = \frac{1}{4n}.$$

To make the probability of being outside the interval at most 0.01, it is enough to require

$$\frac{1}{4n} \leq 0.01.$$

Thus $n \geq 25$ is sufficient. With $n = 25$, Chebyshev gives

$$P(|\bar{X}_n - \mu| < 2\sigma) \geq 0.99.$$

MATH 394: Probability I

Homework 8 Solutions

Problem 5.

Source: Blitzstein and Hwang, *Introduction to Probability*, Chapter 10, Problem 4.

The famous arithmetic mean–geometric mean inequality says that for any positive numbers

$$a_1, a_2, \dots, a_n,$$

we have

$$\frac{a_1 + a_2 + \dots + a_n}{n} \geq (a_1 a_2 \dots a_n)^{1/n}.$$

Show that this inequality follows from Jensen's inequality, by considering $E(\log X)$ for a random variable X whose possible values are $a_1, \dots, a_n$. You should specify the PMF of X .

Solution

Let X take the values $a_1, a_2, \dots, a_n$ with equal probabilities:

$$P(X = a_j) = \frac{1}{n}, \quad j = 1, \dots, n.$$

Then

$$E(X) = \frac{a_1 + \dots + a_n}{n}$$

and

$$E(\log X) = \frac{1}{n} \sum_{j=1}^n \log a_j = \log \left((a_1 a_2 \dots a_n)^{1/n} \right).$$

Since $\log x$ is concave, Jensen's inequality gives

$$E(\log X) \leq \log(E(X)).$$

Therefore

$$\log \left((a_1 a_2 \dots a_n)^{1/n} \right) \leq \log \left(\frac{a_1 + \dots + a_n}{n} \right).$$

Because the logarithm is increasing, Exponentiating both sides gives

$$(a_1 a_2 \dots a_n)^{1/n} \leq \frac{a_1 + a_2 + \dots + a_n}{n}.$$

MATH 394: Probability I

Homework 8 Solutions

Problem 6.

Source: Blitzstein and Hwang, *Introduction to Probability*, Chapter 10, Problem 7.

Let X and Y be i.i.d. positive random variables, and let $c > 0$. For each part below, fill in the appropriate equality or inequality symbol.

Write $=$ if the two sides are always equal, $\leq$ if the left-hand side is less than or equal to the right-hand side but not necessarily equal, and similarly for $\geq$. If no relation holds in general, write $?$.

- (a) $E(\log X) \text{ ______ } \log(EX)$
- (b) $E(X) \text{ ______ } \sqrt{E(X^2)}$
- (c) $E(\sin^2 X) + E(\cos^2 X) \text{ ______ } 1$
- (d) $E(|X|) \text{ ______ } \sqrt{E(X^2)}$
- (e) $P(X > c) \text{ ______ } \frac{E(X^3)}{c^3}$
- (f) $P(X \leq Y) \text{ ______ } P(X \geq Y)$
- (g) $E(XY) \text{ ______ } \sqrt{E(X^2)E(Y^2)}$
- (h) $P(X + Y > 10) \text{ ______ } P(X > 5 \text{ or } Y > 5)$
- (i) $E(\min(X, Y)) \text{ ______ } \min(EX, EY)$
- (j) $E(X/Y) \text{ ______ } \frac{EX}{EY}$
- (k) $E(X^2(X^2 + 1)) \text{ ______ } E(X^2(Y^2 + 1))$
- (l) $E\left(\frac{X^3}{X^3 + Y^3}\right) \text{ ______ } E\left(\frac{Y^3}{X^3 + Y^3}\right)$

Solution

The answers are as follows.

Part	Symbol	Reason
(a)	$\leq$	$\log x$ is concave, by Jensen.
(b)	$\leq$	$(EX)^2 \leq E(X^2)$.
(c)	$=$	$\sin^2 X + \cos^2 X = 1$ pointwise.
(d)	$\leq$	$(E X)^2 \leq E(X^2)$.
(e)	$\leq$	Markov applied to X^3 .
(f)	$=$	Symmetry of i.i.d. X, Y .
(g)	$\leq$	Cauchy–Schwarz.
(h)	$\leq$	$X + Y > 10$ implies $X > 5$ or $Y > 5$.
(i)	$\leq$	$\min(X, Y) \leq X$ and $\min(X, Y) \leq Y$.
(j)	$?$	No fixed relation holds in general.
(k)	$\geq$	$E(X^4) \geq (EX^2)^2$.
(l)	$=$	Symmetry, and the two fractions exchange when X, Y are swapped.

MATH 394: Probability I**Homework 8 Solutions**

For part (k), independence gives

$$E(X^2(Y^2 + 1)) = (EX^2)^2 + E(X^2),$$

while the left-hand side is $E(X^4) + E(X^2)$, so the comparison reduces to $E(X^4) \geq (EX^2)^2$.

MATH 394: Probability I

Homework 8 Solutions

Problem 7.

Source: Blitzstein and Hwang, *Introduction to Probability*, Chapter 10, Problem 22.

Let

$$U_1, U_2, \dots, U_{60}$$

be i.i.d. $\text{Unif}(0, 1)$ and let

$$X = U_1 + U_2 + \dots + U_{60}.$$

- (a) Which important distribution is the distribution of X very close to? Specify what the parameters are, and state which theorem justifies your choice.
- (b) Give a simple but accurate approximation for

$$P(X > 17).$$

Justify briefly.

Solution

(a) For one $\text{Unif}(0, 1)$ random variable,

$$E(U_j) = \frac{1}{2}, \quad \text{Var}(U_j) = \frac{1}{12}.$$

Therefore

$$E(X) = 60 \cdot \frac{1}{2} = 30, \quad \text{Var}(X) = 60 \cdot \frac{1}{12} = 5.$$

By the Central Limit Theorem, the distribution of X is very close to

$$N(30, 5).$$

(b) Using the normal approximation,

$$P(X > 17) \approx P(N(30, 5) > 17) = P\left(Z > \frac{17 - 30}{\sqrt{5}}\right).$$

The standardized value is

$$\frac{17 - 30}{\sqrt{5}} \approx -5.81.$$

Thus

$$P(X > 17) \approx P(Z > -5.81) = \Phi(5.81),$$

which is essentially 1.

MATH 394: Probability I

Homework 8 Solutions

Problem 8.

Source: Blitzstein and Hwang, *Introduction to Probability*, Chapter 10, Problem 25.

(a) Let

$$Y = e^X, \quad X \sim \text{Exp}(3).$$

Find the mean and variance of Y .

(b) For $Y_1, \dots, Y_n$ i.i.d. with the same distribution as Y from part (a), what is the approximate distribution of the sample mean

$$\bar{Y}_n = \frac{1}{n} \sum_{j=1}^n Y_j$$

when n is large?

Solution

(a) Since $X \sim \text{Exp}(3)$, its MGF is

$$M_X(t) = \frac{3}{3-t}, \quad t < 3.$$

Because $Y = e^X$,

$$E(Y) = E(e^X) = M_X(1) = \frac{3}{2}.$$

Also

$$E(Y^2) = E(e^{2X}) = M_X(2) = 3.$$

Therefore

$$\text{Var}(Y) = E(Y^2) - [E(Y)]^2 = 3 - \frac{9}{4} = \frac{3}{4}.$$

(b) By the Central Limit Theorem, when n is large,

$$\bar{Y}_n \approx N\left(E(Y), \frac{\text{Var}(Y)}{n}\right) = N\left(\frac{3}{2}, \frac{3}{4n}\right).$$

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Homework 8 Solutions

Problem 9.

Source: Blitzstein and Hwang, *Introduction to Probability*, Chapter 10, Problem 27.

Consider i.i.d. $\text{Pois}(\lambda)$ random variables

$$X_1, X_2, \dots$$

The MGF of X_j is

$$M(t) = e^{\lambda(e^t - 1)}.$$

(a) Find the MGF $M_n(t)$ of the sample mean

$$\bar{X}_n = \frac{1}{n} \sum_{j=1}^n X_j.$$

(b) Find the limit of $M_n(t)$ as $n \rightarrow \infty$. You can do this with almost no calculation using a relevant theorem, or you can use part (a) and the fact that $e^x \approx 1 + x$ if x is very small.

Solution

(a) We compute

$$M_n(t) = E\left(e^{t\bar{X}_n}\right) = E\left(e^{(t/n)(X_1 + \dots + X_n)}\right).$$

By independence,

$$M_n(t) = \prod_{j=1}^n E\left(e^{(t/n)X_j}\right) = \left[M\left(\frac{t}{n}\right)\right]^n.$$

Using the Poisson MGF,

$$M_n(t) = \left[e^{\lambda(e^{t/n} - 1)}\right]^n = \exp\left(n\lambda(e^{t/n} - 1)\right).$$

(b) As $n \rightarrow \infty$,

$$e^{t/n} - 1 \sim \frac{t}{n}.$$

Therefore

$$n\lambda(e^{t/n} - 1) \rightarrow \lambda t.$$

It follows that

$$M_n(t) \rightarrow e^{\lambda t}.$$

This is the MGF of the constant random variable equal to λ , which agrees with the Law of Large Numbers.

University of Washington
Department of Mathematics

Midterm Examination Solutions

MATH 394: Probability I

Summer 2026 — Instructor: Arman Jahangiri

Exam date: July 22, 2026
Answering time: 60 minutes
Total: 60 points

MATH 394: Probability I

Midterm Solutions

Problem 1.

5 points

A student organization has 7 mathematics graduate students, 5 statistics graduate students, and 6 undergraduate students. A committee of 5 students is selected uniformly at random from all committees of 5.

Find the probability that the committee contains exactly 2 mathematics graduate students, exactly 1 statistics graduate student, and exactly 2 undergraduate students.

Give an exact answer. A correct expression involving binomial coefficients is acceptable.

Solution

There are

$$\binom{18}{5}$$

possible committees of 5 students, and all are equally likely.

To form a committee of the required type, choose:

- 2 of the 7 mathematics graduate students,
- 1 of the 5 statistics graduate students, and
- 2 of the 6 undergraduate students.

Thus the number of favorable committees is

$$\binom{7}{2} \binom{5}{1} \binom{6}{2}.$$

Therefore,

$$P(\text{required composition}) = \frac{\binom{7}{2} \binom{5}{1} \binom{6}{2}}{\binom{18}{5}}.$$

Numerically,

$$\binom{7}{2} \binom{5}{1} \binom{6}{2} = 21 \cdot 5 \cdot 15 = 1575, \quad \binom{18}{5} = 8568,$$

so

$$\frac{1575}{8568} = \frac{525}{2856} = \frac{175}{952}.$$

MATH 394: Probability I

Midterm Solutions

Problem 2.

10 points

A spam filter is designed by looking at commonly occurring phrases in spam. Suppose that 80% of email is spam. In 10% of the spam emails, the phrase “free money” is used, whereas this phrase is only used in 1% of non-spam emails. A new email has just arrived, which does mention “free money.” What is the probability that it is spam?

Solution

(a) Let S be the event that the email is spam, and let F be the event that the email contains the phrase “free money.” We are given

$$P(S) = 0.80, \quad P(F | S) = 0.10, \quad P(F | S^c) = 0.01.$$

Bayes’ rule gives

$$P(S | F) = \frac{P(F | S)P(S)}{P(F | S)P(S) + P(F | S^c)P(S^c)}.$$

Substituting the given values,

$$P(S | F) = \frac{0.10 \cdot 0.80}{0.10 \cdot 0.80 + 0.01 \cdot 0.20} = \frac{0.08}{0.082} = \frac{40}{41}.$$

MATH 394: Probability I

Midterm Solutions

Problem 3.

10 points

Let A and B be events.

- (a) Let S be a sample space and $A_1, \dots, A_n \in \mathcal{F}$ be events. Explain what it means for A_i 's to form a partition of S , **(3 points)**
- (b) Suppose $P(A) = 0.6$ and $P(B) = 0.5$. Can A and B be disjoint? **(3 points)**
- (c) Suppose that A is independent of itself. Use the definition of independence to conclude that $P(A) = 0$ or $P(A) = 1$. **(4 points)**

Solution

- (a) The events $A_1, \dots, A_n$ form a partition of the sample space S if they are pairwise disjoint,

$$A_i \cap A_j = \emptyset, \quad i \neq j,$$

and

$$\bigcup_{i=1}^n A_i = S.$$

- (b) If A and B were disjoint, then

$$1 \geq P(A \cup B) = P(A) + P(B) = 0.6 + 0.5 = 1.1,$$

which is impossible since probabilities cannot exceed 1. Hence A and B cannot be disjoint.

- (c) If A is independent of itself, then

$$P(A \cap A) = P(A)P(A).$$

Since $A \cap A = A$,

$$P(A) = P(A)^2.$$

Let $p = P(A)$. Then

$$p(1 - p) = 0,$$

so $p = 0$ or $p = 1$.

Final Answer

- (a) $\{A_i\}_{i=1}^n$ is a partition iff the events are pairwise disjoint and their union is S .
 (b) No.
 (c)

$$P(A) = 0 \quad \text{or} \quad P(A) = 1.$$

MATH 394: Probability I

Midterm Solutions

Problem 4.

15 points

Suppose that X has probability density function

$$f_X(x) = \begin{cases} cx(2-x), & 0 < x < 2, \\ 0, & \text{otherwise,} \end{cases}$$

where c is a constant.

- (a) Find c . (3 points)
- (b) Find the cumulative distribution function $F_X(x)$ for all real x . (5 points)
- (c) Find $P(X > 3/2)$. (2 points)
- (d) Find $\mathbb{E}[X]$ and $\text{Var}(X)$. (5 points)

Solution

(a) A density must integrate to 1, so

$$\begin{aligned} 1 &= c \int_0^2 x(2-x) dx \\ &= c \int_0^2 (2x - x^2) dx \\ &= c \left[x^2 - \frac{x^3}{3} \right]_0^2 = c \left(4 - \frac{8}{3} \right) = c \frac{4}{3}. \end{aligned}$$

Hence

$$c = \frac{3}{4}.$$

(b) For $0 < x < 2$,

$$\begin{aligned} F_X(x) &= \int_0^x \frac{3}{4} t(2-t) dt \\ &= \frac{3}{4} \left[t^2 - \frac{t^3}{3} \right]_0^x \\ &= \frac{3x^2}{4} - \frac{x^3}{4}. \end{aligned}$$

Therefore,

$$F_X(x) = \begin{cases} 0, & x \leq 0, \\ \frac{3x^2 - x^3}{4}, & 0 < x < 2, \\ 1, & x \geq 2. \end{cases}$$

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(c) Using the CDF,

$$\begin{aligned} P\left(X > \frac{3}{2}\right) &= 1 - F_X\left(\frac{3}{2}\right) \\ &= 1 - \frac{3(3/2)^2 - (3/2)^3}{4} \\ &= 1 - \frac{27}{32} = \frac{5}{32}. \end{aligned}$$

(d) First,

$$\begin{aligned} \mathbb{E}[X] &= \int_0^2 x \frac{3}{4} x(2-x) dx \\ &= \frac{3}{4} \int_0^2 (2x^2 - x^3) dx \\ &= \frac{3}{4} \left[\frac{2x^3}{3} - \frac{x^4}{4} \right]_0^2 = 1. \end{aligned}$$

Also,

$$\begin{aligned} \mathbb{E}[X^2] &= \int_0^2 x^2 \frac{3}{4} x(2-x) dx \\ &= \frac{3}{4} \int_0^2 (2x^3 - x^4) dx \\ &= \frac{3}{4} \left[\frac{x^4}{2} - \frac{x^5}{5} \right]_0^2 \\ &= \frac{6}{5}. \end{aligned}$$

Thus

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{6}{5} - 1 = \frac{1}{5}.$$

Final Answer

$$c = \frac{3}{4}, \quad P(X > 3/2) = \frac{5}{32}, \quad \mathbb{E}[X] = 1, \quad \text{Var}(X) = \frac{1}{5}.$$

The CDF is

$$F_X(x) = \begin{cases} 0, & x \leq 0, \\ (3x^2 - x^3)/4, & 0 < x < 2, \\ 1, & x \geq 2. \end{cases}$$

MATH 394: Probability I

Midterm Solutions

Problem 5.

15 points

A company has found that 40% of customers who visit its website make a purchase. Suppose that 200 customers visit the website independently on a particular day.

Let X denote the number of customers who make a purchase.

1. Determine whether a **Poisson approximation** or a **normal approximation** is more appropriate by verifying the required conditions. Then clearly state the approximating distribution; that is, provide λ if a Poisson approximation is appropriate, or provide μ and σ^2 if a normal approximation is appropriate. (5 points)

2. Use the approximation from part (1) to estimate each of the following: (10 points)

(a)

$$\mathbb{P}(70 < X < 90)$$

(3 points)

(b)

$$\mathbb{P}(X > 90)$$

(3 points)

(c)

$$\mathbb{P}(X = 80)$$

(4 points)

Solution

Since each customer independently makes a purchase with probability $p = 0.40$, we have

$$X \sim \text{Bin}(200, 0.40).$$

(1) For a Poisson approximation to be appropriate, n should be large and p should be small. Although

$$n = 200$$

is large, $p = 0.40$ is not small. Therefore, a Poisson approximation is not appropriate.

For a normal approximation, we verify that

$$np = 200(0.40) = 80$$

and

$$n(1 - p) = 200(0.60) = 120.$$

Since both quantities are greater than 10, the normal approximation is appropriate.

The mean and variance of X are

$$\mu = np = 80$$

and

$$\sigma^2 = np(1 - p) = 200(0.40)(0.60) = 48.$$

Therefore, we approximate X by

$$Y \sim N(80, 48),$$

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where

$$\sigma = \sqrt{48} \approx 6.93.$$

(2)

(a) Using the continuity correction,

$$\mathbb{P}(70 < X < 90) \approx \mathbb{P}(70.5 < Y < 89.5).$$

Standardizing,

$$\begin{aligned} \mathbb{P}(70.5 < Y < 89.5) &= \mathbb{P}\left(\frac{70.5 - 80}{\sqrt{48}} < Z < \frac{89.5 - 80}{\sqrt{48}}\right) \\ &\approx \mathbb{P}(-1.37 < Z < 1.37). \end{aligned}$$

Using the standard normal table,

$$\Phi(1.37) = 0.9147$$

and

$$\Phi(-1.37) = 1 - \Phi(1.37) = 0.0853.$$

Therefore,

$$\mathbb{P}(70 < X < 90) \approx 0.9147 - 0.0853 = 0.8294.$$

(b) Using the continuity correction,

$$\mathbb{P}(X > 90) \approx \mathbb{P}(Y > 90.5).$$

Standardizing,

$$\begin{aligned} \mathbb{P}(Y > 90.5) &= \mathbb{P}\left(Z > \frac{90.5 - 80}{\sqrt{48}}\right) \\ &\approx \mathbb{P}(Z > 1.52). \end{aligned}$$

Using the standard normal table,

$$\Phi(1.52) = 0.9357.$$

Thus,

$$\mathbb{P}(X > 90) \approx 1 - \Phi(1.52) = 1 - 0.9357 = 0.0643.$$

(c) Using the continuity correction,

$$\mathbb{P}(X = 80) \approx \mathbb{P}(79.5 < Y < 80.5).$$

Standardizing,

$$\begin{aligned} \mathbb{P}(79.5 < Y < 80.5) &= \mathbb{P}\left(\frac{79.5 - 80}{\sqrt{48}} < Z < \frac{80.5 - 80}{\sqrt{48}}\right) \\ &\approx \mathbb{P}(-0.07 < Z < 0.07). \end{aligned}$$

Using the standard normal table,

$$\Phi(0.07) = 0.5279$$

and

$$\Phi(-0.07) = 0.4721.$$

Therefore,

$$\mathbb{P}(X = 80) \approx 0.5279 - 0.4721 = 0.0558.$$

University of Washington
Department of Mathematics

Final Examination
MATH 394: Probability I

Summer 2026 — Instructor: Arman Jahangiri

Date:

August 21, 2026

Answering time:

60 minutes

Total points:

60

Name:

Student ID:

Signature:

Instructions

- This examination contains **seven problems**. Verify that your copy is complete.
- Show all work and justify your answers. Unsupported answers may receive little or no credit.
- Clearly define any events or random variables that you introduce.
- The estimated times are provided only as a guide for pacing.
- Do not remove pages from the examination booklet.

Permitted materials

Examinations are closed-book and closed-notes. Students may use:

- one handwritten 8.5×11 inch reference sheet (front and back), and
- a non-programmable non-graphing calculator capable of standard exponential and logarithmic computations (e.g. TI-30X IIS).

No other electronic devices, calculators, communication devices, or external resources may be used during examinations.

Problem	Score
1	/16
2	/8
3	/7
4	/7
5	/8
6	/8
7	/6
Total	/60

By signing above, you affirm that the work is your own and that you followed the examination rules.

MATH 394: Probability I**Final Examination****Problem 1.****Estimated time: 13 min****16 points**

Let the joint probability density function of random variables X and Y be given by

$$f(x, y) = \begin{cases} x^2 e^{-x(y+1)}, & x \geq 0, \quad y \geq 0, \\ 0, & \text{elsewhere.} \end{cases}$$

The marginal probability density functions are

$$f_X(x) = \begin{cases} x e^{-x}, & x \geq 0, \\ 0, & \text{otherwise,} \end{cases}$$

and

$$f_Y(y) = \begin{cases} \frac{2}{(1+y)^3}, & y \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Calculate

$$\text{Cov}(X, Y).$$

(4 points)

(b) Calculate

$$\text{Cov}(3X + 1, 2X - 4).$$

(3 points)

(c) Are X and Y independent? Justify your answer.

(2 points)

(d) Calculate the moment-generating function of X , namely

$$M_X(t) = E(e^{tX}).$$

(5 points)

(e) Using $M_X(t)$, calculate $E(X)$.

(2 points)

MATH 394: Probability I**Final Examination****Problem 2.****Estimated time: 9 min****8 points**

Let X and Y be two positive independent continuous random variables with probability density functions $f_1(x)$ and $f_2(y)$, respectively.

Find the probability density function of

$$U = \frac{X}{Y}.$$

Hint: Let $V = X$. Find the joint probability density function of U and V (Jacobian method), and then calculate the marginal probability density function of U .

MATH 394: Probability I**Final Examination****Problem 3.****Estimated time: 5 min****7 points**

Let X and Y be i.i.d. positive random variables. Assume that all expectations appearing below exist and are finite.

For each part below, fill in the appropriate equality or inequality symbol.

Write $=$ if the two sides are always equal, $\leq$ if the left-hand side is less than or equal to the right-hand side but not necessarily equal, and similarly for $\geq$. If no relation holds in general, write $?$.

Justify each answer.

(a)

$$E(\log X) \quad ______ \quad \log(E(X)).$$

(3 points)

(b)

$$P(X \leq Y) \quad ______ \quad P(X \geq Y).$$

(2 points)

(c)

$$E(XY) \quad ______ \quad \sqrt{E(X^2)E(Y^2)}.$$

(2 points)

MATH 394: Probability I**Final Examination****Problem 4.****Estimated time: 8 min****7 points**

Suppose

$$X_1, \dots, X_n$$

are independent Bernoulli random variables with unknown success probability p , and define

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Let

$$\varepsilon > 0 \quad \text{and} \quad 0 < \alpha < 1.$$

Using Chebyshev's inequality, derive a condition on the sample size n that guarantees

$$P(|\hat{p} - p| < \varepsilon) \geq 1 - \alpha$$

for every

$$0 < p < 1.$$

MATH 394: Probability I**Final Examination****Problem 5.****Estimated time: 7 min****8 points**

Let X and Y be independent exponential random variables with common parameter $\lambda > 0$:

$$X, Y \stackrel{\text{i.i.d.}}{\sim} \text{Exp}(\lambda),$$

with probability density function

$$f(w) = \begin{cases} \lambda e^{-\lambda w}, & w > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Define

$$T = X + Y.$$

Using the convolution formula

$$f_T(t) = \int_{-\infty}^{\infty} f_X(x)f_Y(t-x) dx,$$

find the probability density function of T , including its support.

MATH 394: Probability I**Final Examination****Problem 6.****Estimated time: 5 min****8 points**

Let X, Y, Z be jointly continuous with joint probability density function

$$f_{X,Y,Z}(x, y, z) = \begin{cases} x^2 e^{-x(1+y+z)}, & x, y, z > 0, \\ 0, & \text{otherwise.} \end{cases}$$

The marginal probability density functions are

$$f_X(x) = \begin{cases} e^{-x}, & x > 0, \\ 0, & \text{otherwise,} \end{cases}$$

and

$$f_Y(y) = \begin{cases} \frac{1}{(1+y)^2}, & y > 0, \\ 0, & \text{otherwise,} \end{cases} \quad f_Z(z) = \begin{cases} \frac{1}{(1+z)^2}, & z > 0, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Are X, Y, Z pairwise independent? Justify your answer. **(5 points)**
- (b) Are X, Y, Z mutually independent? Justify your answer. **(3 points)**

MATH 394: Probability I

Final Examination

Problem 7.

Estimated time: 4 min

6 points

For each part below, select the **correct statement** of the indicated theorem. Only one choice is correct in each part.

(a) **Weak Law of Large Numbers (WLLN).**

Let $X_1, X_2, \dots$ be i.i.d. random variables with finite mean

$$E(X_i) = \mu$$

and finite variance

$$\text{Var}(X_i) = \sigma^2 > 0.$$

Let

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Which of the following is the conclusion of the Weak Law of Large Numbers?

(2 points)

(A) For every $\varepsilon > 0$,

$$P(|\bar{X}_n - \mu| > \varepsilon) = 0 \quad \text{for every } n.$$

(B) For every $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| > \varepsilon) = 0.$$

(C)

$$P\left(\lim_{n \rightarrow \infty} \bar{X}_n = \mu\right) = 1.$$

(D) For every $z \in \mathbb{R}$,

$$\lim_{n \rightarrow \infty} P\left(\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \leq z\right) = \Phi(z).$$

(b) **Strong Law of Large Numbers (SLLN).**

Let $X_1, X_2, \dots$ be i.i.d. random variables satisfying the same assumptions as in the Weak Law of Large Numbers, with

$$E(X_i) = \mu$$

and finite variance

$$\text{Var}(X_i) = \sigma^2 > 0.$$

Which of the following is the conclusion of the Strong Law of Large Numbers?

(2 points)

(A)

$$P\left(\lim_{n \rightarrow \infty} \bar{X}_n = \mu\right) = 1.$$

(B) For every $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| > \varepsilon) = 0.$$

(C)

$$\lim_{n \rightarrow \infty} P(\bar{X}_n = \mu) = 1.$$

(D)

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1).$$

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MATH 394: Probability I**Final Examination**

(c) Central Limit Theorem (CLT).

Let $X_1, X_2, \dots$ be i.i.d. random variables with

$$E(X_i) = \mu, \quad \text{Var}(X_i) = 0 < \sigma^2 < \infty.$$

Which of the following is the conclusion of the Central Limit Theorem?

(2 points)

(A)

$$\bar{X}_n \xrightarrow{P} N(\mu, \sigma^2).$$

(B)

$$P\left(\lim_{n \rightarrow \infty} \bar{X}_n = \mu\right) = 1.$$

(C) For every $z \in \mathbb{R}$,

$$\lim_{n \rightarrow \infty} P\left(\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \leq z\right) = \Phi(z).$$

(D) For every $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} P(|\bar{X}_n - \mu| > \varepsilon) = 0.$$

University of Washington
Department of Mathematics

Mini-Project

MATH 394: Probability I

Instructor: Arman Jahangiri

Submission: Upload a PDF of your slides and either a working video link
or an mp4 file on Canvas

Deadline: Friday, August 14, 2026 at 10:00 PM Pacific Time

This mini-project is worth **2% bonus on top of the final course grade**.
This is an **individual project**. The goal is to practice academic presentation and
mathematical communication.

MATH 394: Probability I**Mini-Project****Overview**

For this mini-project, you will create a short recorded presentation in which you teach a probabilistic concept **not covered in our course**. You should imagine that you are teaching the topic to a classmate in MATH 394 who has the background of this course.

Project Requirements

Please follow all of the requirements below:

1. **Length.** Your video must be **at least 10 minutes** long.
2. **Individual work.** This is an **individual** project.
3. **Slides are required.** You should prepare slides and present by speaking over your slides.
4. **Presentation style.** Your presentation should mainly consist of **slides plus verbal explanation**. You should **not** rely on writing extensively on the slides while presenting. Brief annotation, circling, underlining, or pointing for clarity is fine.
5. **Recording method.** If you use macOS, I recommend recording with **QuickTime Player**. If you use another operating system, you may use any similar screen-recording app that allows you to record your screen and your voice. The recording should be of high quality and clear. Using cameras to externally record your screen is not accepted.
6. **Slides.** There is no restriction on the number of slides.
7. **Content.** Your presentation should include:
 - a short introduction and motivation,
 - a clear explanation of the main concept,
 - at least one worked example, application, or illustrative scenario,
 - a brief concluding summary.
 - citations
8. **Sources.** Your final slide should list the sources you used. Please use reliable sources and cite them clearly.

Topic Selection

You must choose a topic that is **not covered in our course**. A list of suggested topics appears below.

- Each student should choose **one** topic.
- Please submit your top five topic preferences (ranked from 1 to 5, where 1 is your most preferred topic and 5 is your least preferred) by completing [this form](#) no later than **July 15, 10:00 PM**. If no response is received by the deadline, it will be assumed that you do not wish to participate in this bonus project. After all responses are collected, you will receive your assigned topic. You can then choose to go with it, or in case it is not of your interest, you can choose a topic of your own not in the list.
- If you would like to present on a different topic, that is fine, but you must get approval first.

Suggested Topics

1. Birthday Paradox
2. The Banach Matchbox Problem
3. Coupon Collector Problem
4. Gambler's Ruin
5. Secretary Problem
6. Buffon's Needle
7. Benford's Law
8. Simpson's Paradox
9. St. Petersburg Paradox
10. Markov Chains
11. Random Walks
12. Poisson Processes
13. Branching Processes
14. Monte Carlo Methods
15. Capture–Recapture Estimation
16. Reservoir Sampling
17. Bloom Filters
18. PageRank
19. Reliability of Systems and Networks
20. Random Graphs (Erdős–Rényi Model)

Submission Instructions

Please submit the following on **Canvas**:

1. a **PDF** of your slides, and
2. either:
 - an **mp4** file, or
 - a **working video link**.

MATH 394: Probability I**Mini-Project**

3. In the text prompt on Canvas, please answer Yes or No to the following two questions:

Question 1: Would you like your slides (not the video) to be used anonymously in the future on the instructor's website?

Questions 2: Would you like your slides (not the video) to be shown anonymously as sample projects to the students taking this course in the future?

Your response should be one of the following: (Yes, Yes), (Yes, No), (No, Yes), (No, No).

Please make sure that:

- your video link works,
- your audio is clear,
- your slides are readable,
- your file names include your name.

Grading

- **Mathematical correctness and understanding** — 40%
- **Clarity and organization of explanation** — 30%
- **Quality of example or application** — 20%
- **Professionalism and completion of instructions** — 10%

A strong project is one that explains one idea **clearly, correctly, and concisely**.

Late Policy

Because final grades are due very shortly after the end of the quarter, **late submissions cannot not be accepted**.

Note

This project is meant to be a useful opportunity to practice **learning by teaching**. Choose a topic that you find interesting, explain it carefully, and aim to teach it in a way that would help one of your classmates understand it.

If you have any questions, please email me at armanjg@uw.edu or post it on Ed discussion.